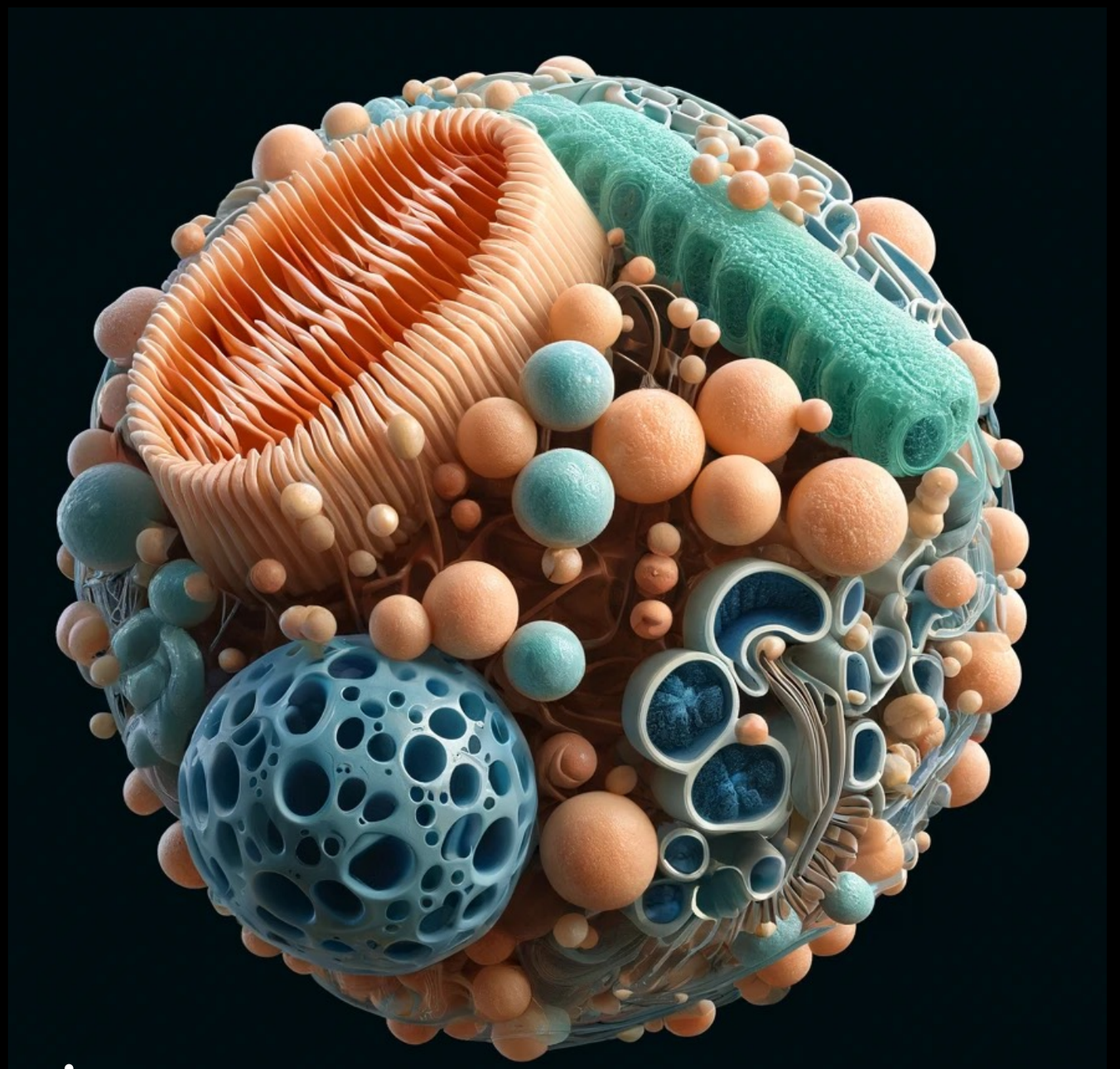
W&M biomath

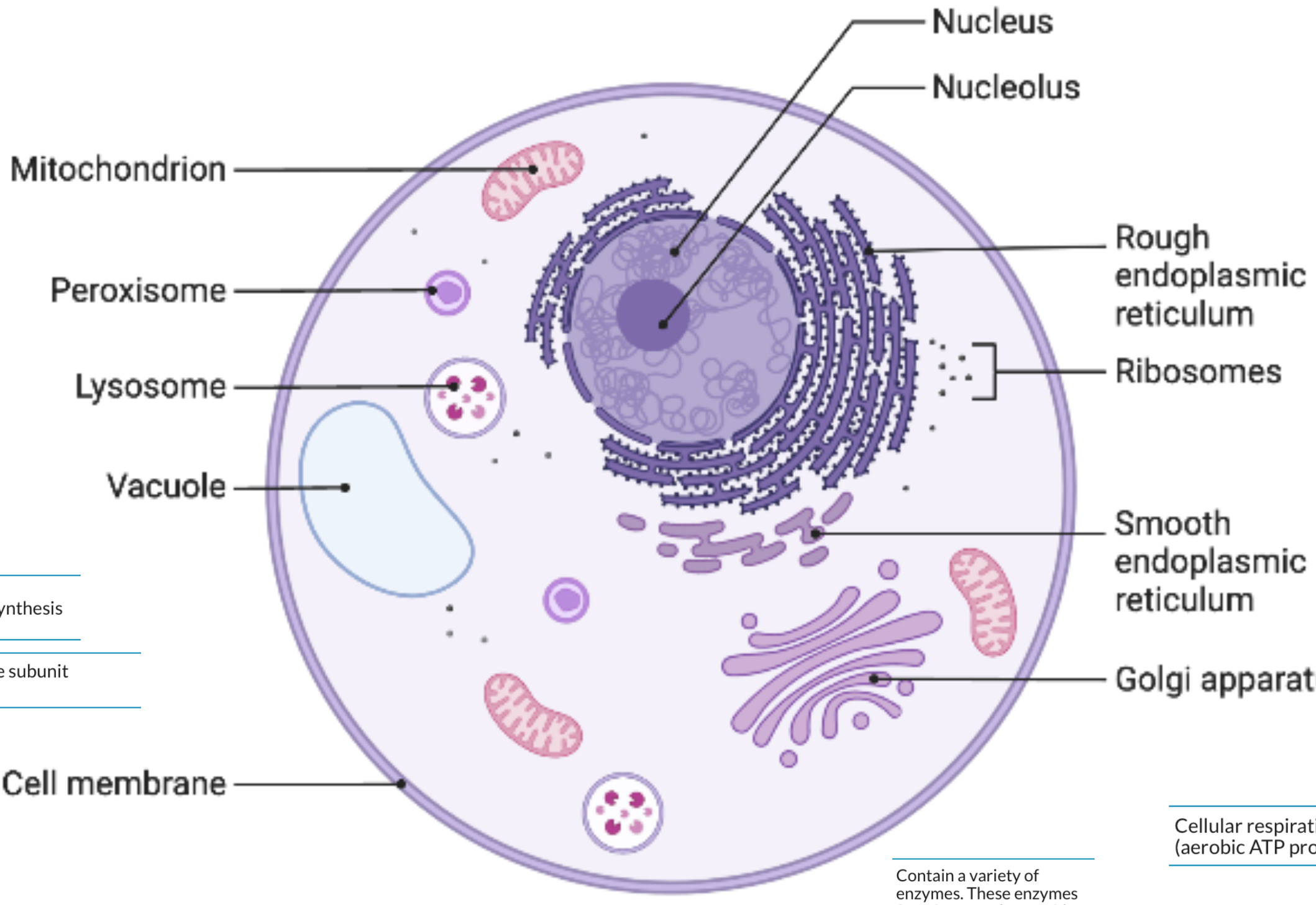


Membraneless organelles, the Cahn-Hilliard equation for liquid-liquid phase separation, and my love-hate relationship with generative AI

Greg Conradi Smith

September 27, 2024

Eukaryotic Cell



Protein synthesis

Ribosome subunit synthesis

Sacs containing digestive enzymes break down captured food materials

Where proteins and lipids coming from the rough ER are processed, sorted, and packaged for transport

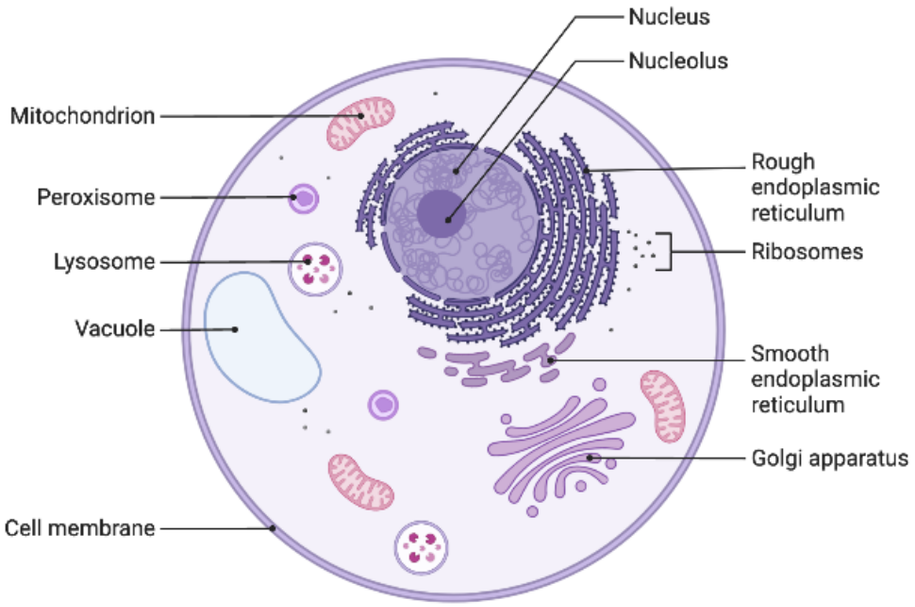
Lipid synthesis

Contain a variety of enzymes. These enzymes function together to rid the cell of toxic substances including hydrogen peroxide a common by-product of cellular respiration.

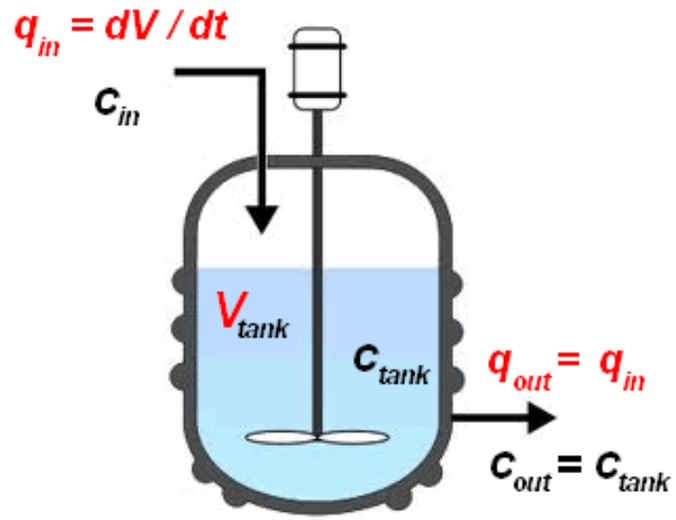
Cellular respiration (aerobic ATP production)

Contains DNA, directs synthesis of proteins and ribosomes

Eukaryotic Cell



continuously stirred tank reactor



spatially heterogenous
compartmentalized
membrane systems

pH
ion concentrations
electrical potential

crowded
energy consuming
non-equilibrium

colloidal / gel-sol like

Molecular crowding and non-thermal activity contribute to the biophysical properties of the cell interior

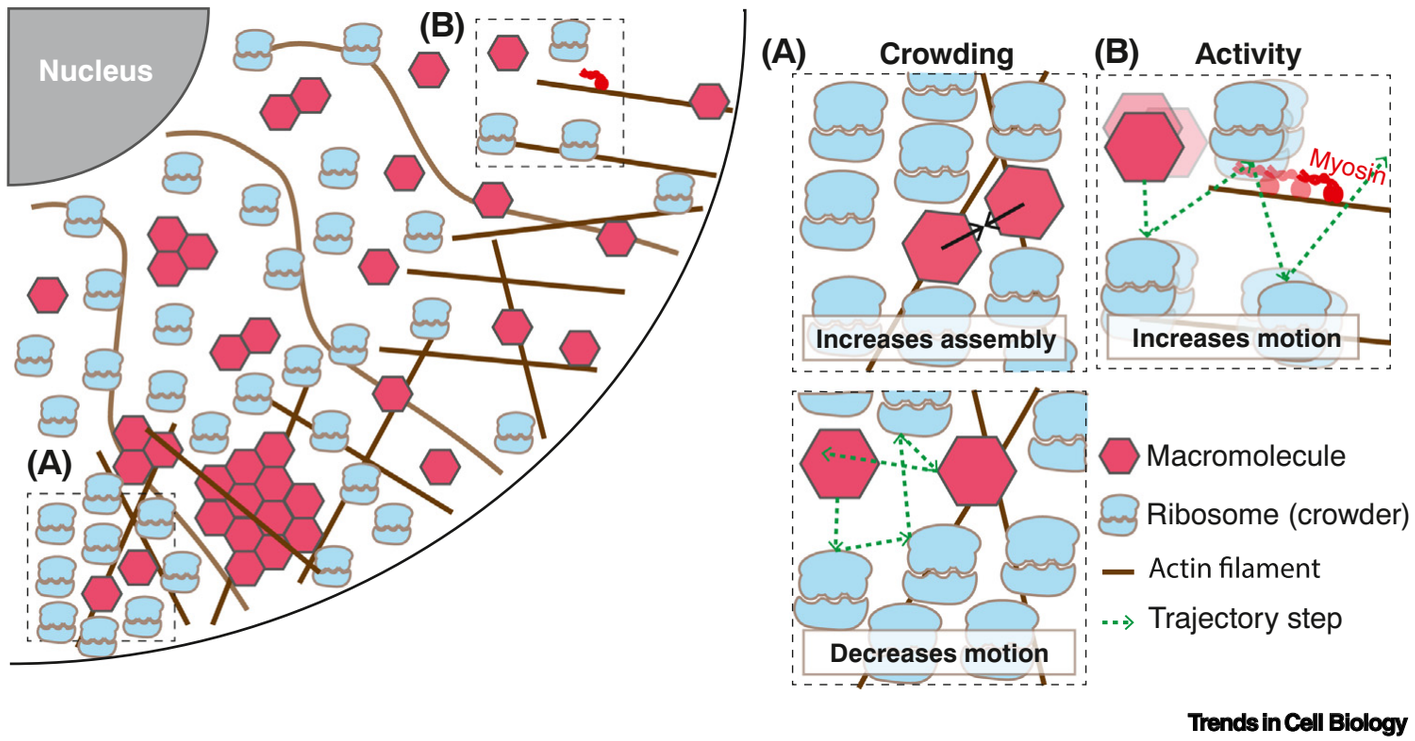
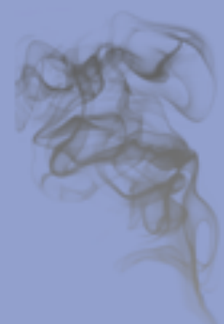


Figure 1. The cell interior is crowded and far from thermal equilibrium. (A) Macromolecular crowding can drive molecular assembly (top) and decrease particle diffusion (bottom). (B) Energy-consuming activities increase particle motion.

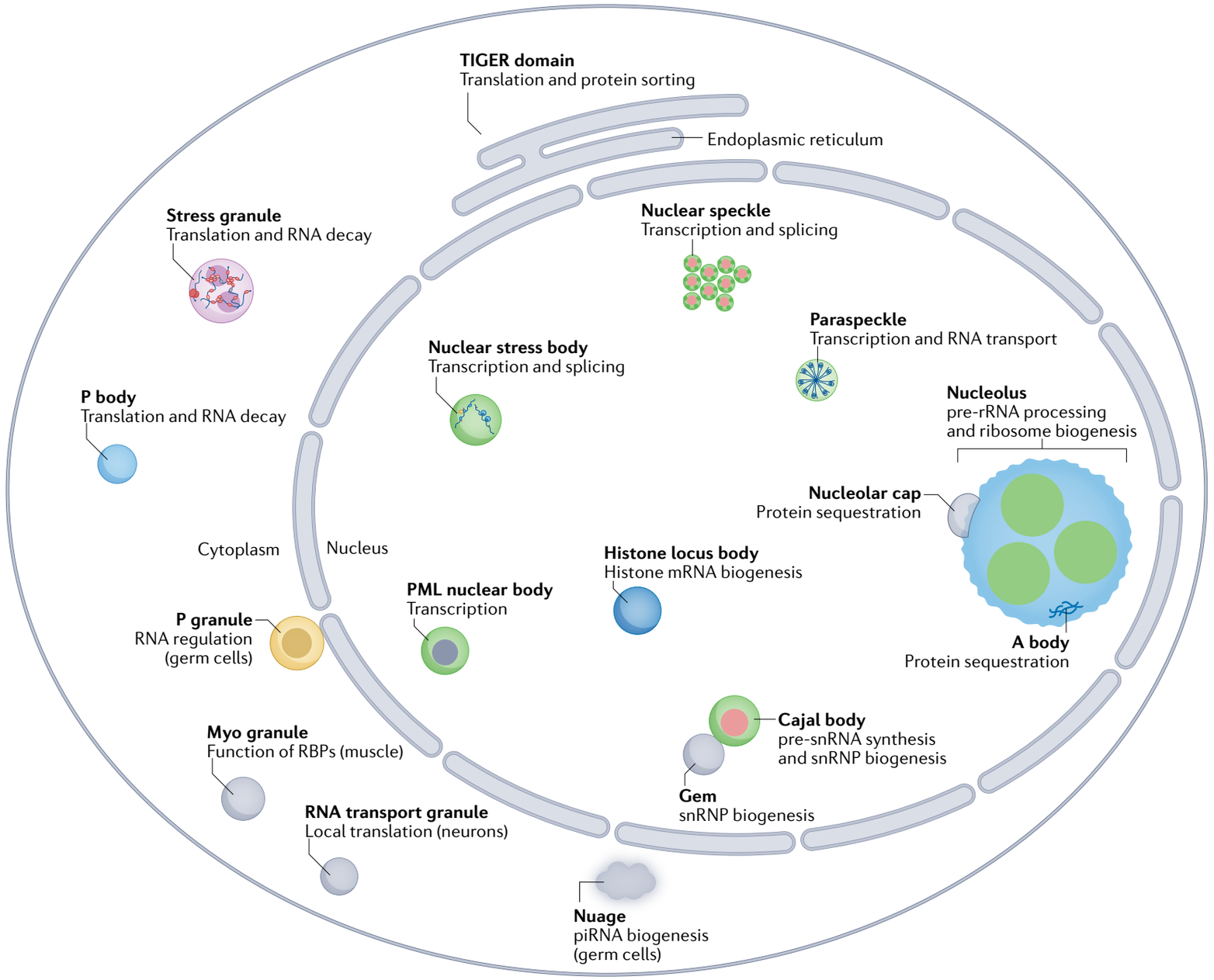
Colloids

a substance which has droplets of one state within a surrounding made up of another state

Droplet \ Surrounding	Solid	Liquid	Gas
Solid	Solid	Sol	 Solid Aerosol
Liquid	Gel	Emulsion	Liquid Aerosol
Gas	Solid Foam	Foam	

https://blogs.ncl.ac.uk/stem/tag/colloid/

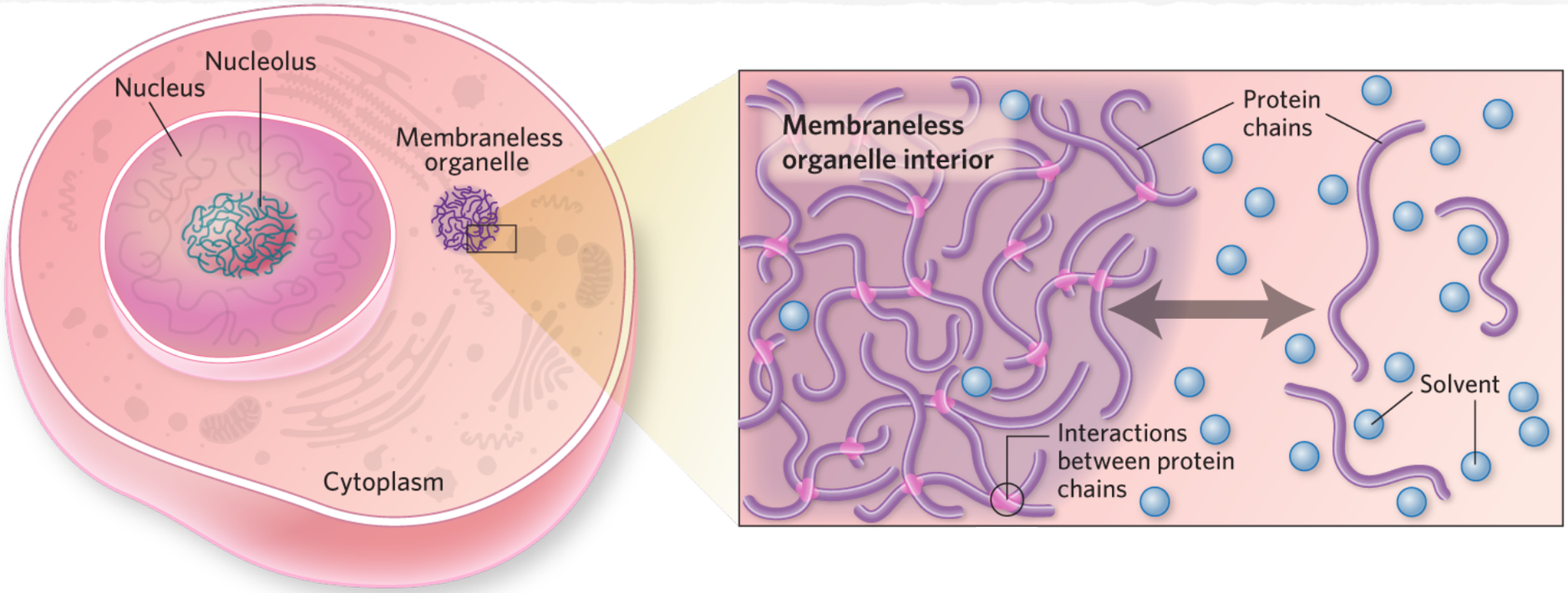
Membraneless organelles in the nucleus and cytoplasm

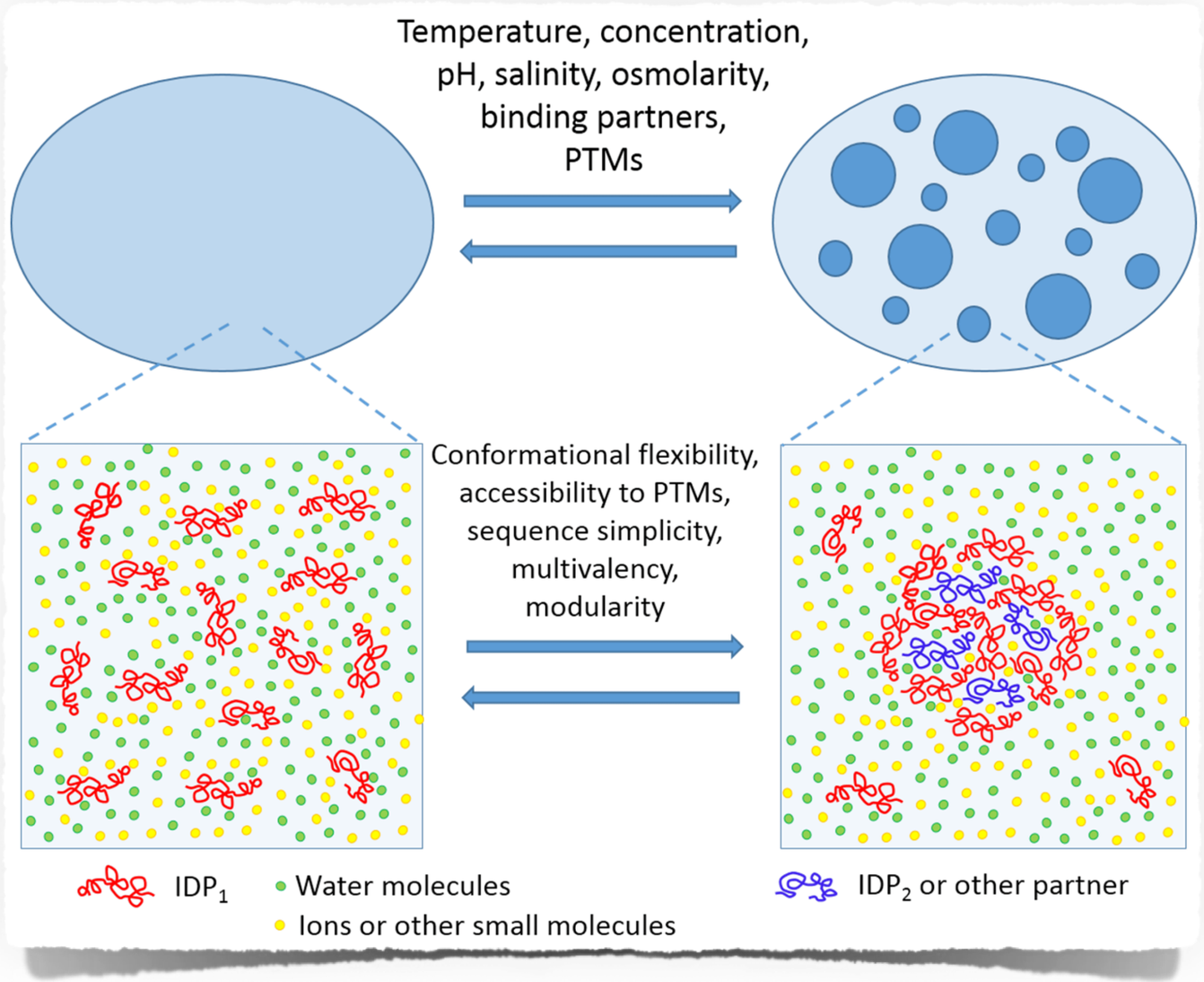


Name (loci involved)	Size (μm)	Number per cell	Gene expression-relevant functions
Nucleolus (rDNA)	0.5–8	2–5	rRNA transcription, processing and modification Ribosome assembly
Cajal body (snRNA)	0.1–2	0–6	snRNA transcription and modification snRNP assembly
Nuclear speckle (multiple)	0.18–1.8	25–50	Transcription Splicing RNA export
PML nuclear body (multiple)	0.1–1	10–30	Transcription Chromatin association
Histone locus body (histone genes)	0.2–1.2	2–4	Histone mRNA transcription and processing
Processing body (NA)	0.4–0.5	0–30	RNA storage and decay, translation
TIGER domain (NA)	Several micrometres	1[a]	Translation and protein sorting
Paraspeckle (*NEAT1* locus with multiple other loci)	0.2–0.5	2–20	Transcription RNA retention
Nuclear stress body (*HSATIII*)	0.3–3	2–10	Transcription Splicing
Stress granule (NA)	0.1–2	5–30	RNA storage and decay Translation

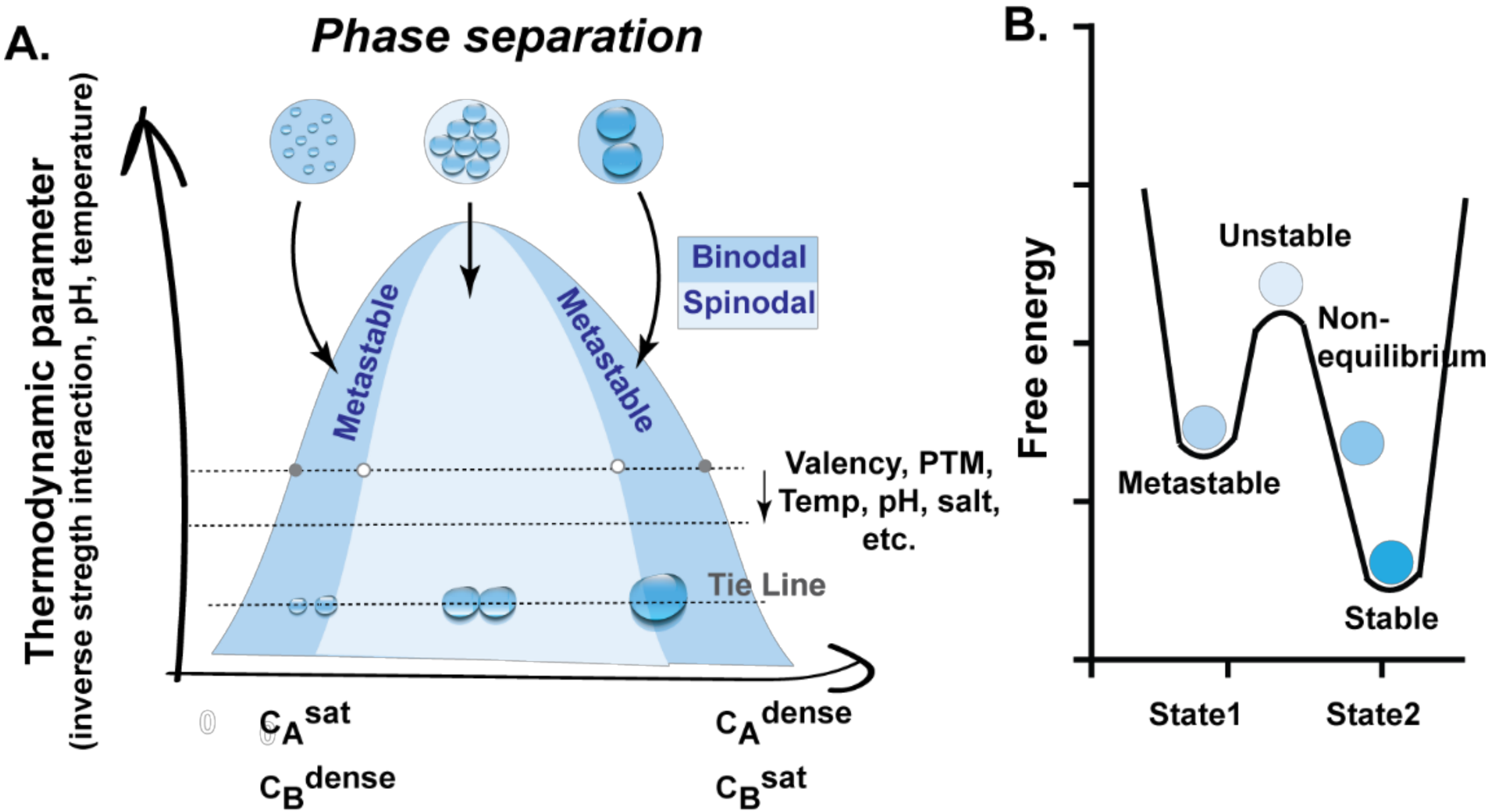
Hirose et al. 2024
Nature Reviews Molecular Cell Biology

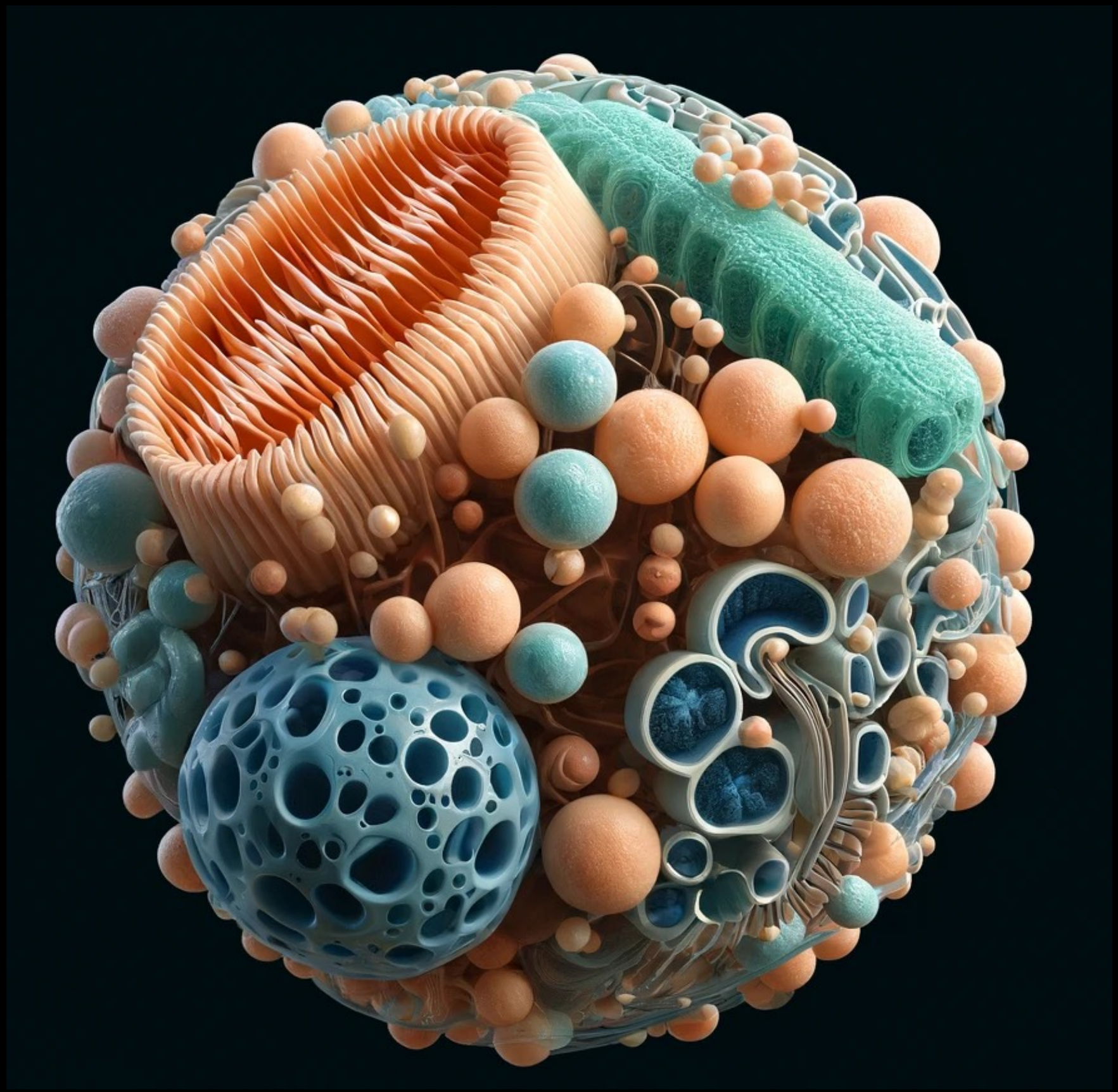
What are membraneless organelles?





Principles of liquid-liquid phase separation (LLPS)





The Cahn-Hilliard equation

The Cahn-Hilliard equation takes the form

$$\frac{\partial \phi}{\partial t} = \nabla \cdot (M(\phi) \nabla \mu)$$

where

- $\phi(\mathbf{x}, t)$ is the order parameter
- $M(\phi)$ is the mobility, a value that can depend on the order parameter
- $\nabla \mu$ is the gradient of the chemical potential $\mu(\mathbf{x}, t)$.

$$\frac{\partial \phi}{\partial t} + \nabla \cdot \mathbf{j} = 0 \qquad \mathbf{j} = -M(\phi) \nabla \mu \qquad \text{What is } \mu(\mathbf{x}, t)?$$

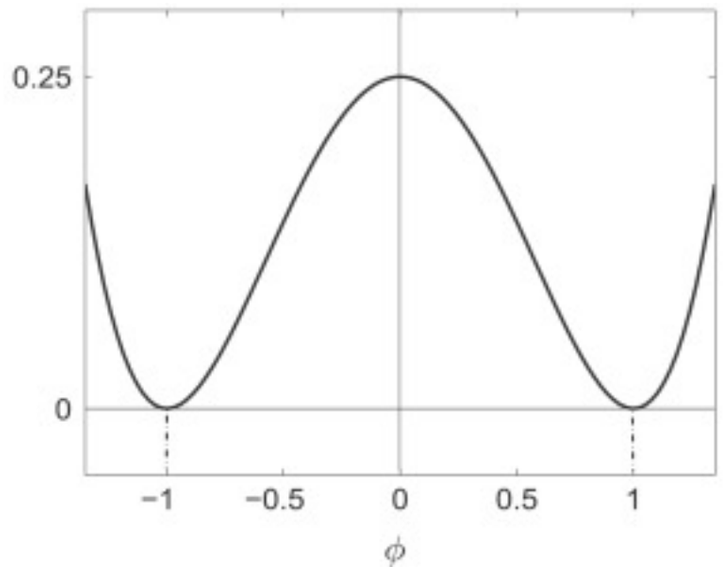
The free energy functional $F[\phi]$ typically has the form:

$$F[\phi] = \int \left(\frac{\sigma}{2} |\nabla \phi|^2 + \frac{1}{4}(\phi^2 - 1)^2 \right) \, d\mathbf{x}$$

penalizes gradients $f(\phi)$ demixing

$\phi(\mathbf{x}, t)$ is the order parameter

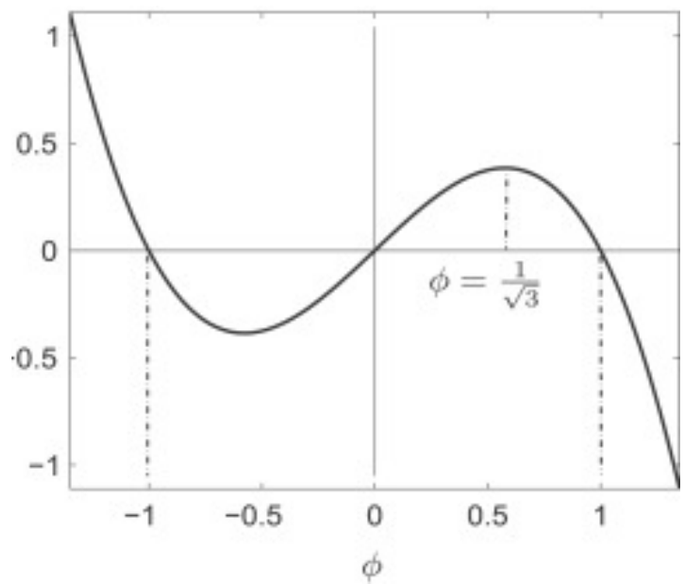
$$f(\phi) = \frac{1}{4}(\phi^2 - 1)^2$$



Using the calculus of variations
the free energy μ is derived from F[ϕ]

$$\mu = \frac{\delta F[\phi]}{\delta \phi} = -\sigma \nabla^2 \phi + (\phi^2 - 1)\phi$$

$$-f'(\phi) = (1 - \phi^2)\phi = \phi - \phi^3$$



Assuming mobility is not function of ϕ

$$\frac{\partial \phi}{\partial t} = M \nabla^2 \left[\phi^3 - \phi - \sigma \nabla^2 \phi \right]$$

$$\nabla = \frac{\partial}{\partial x}\mathbf{e}_x + \frac{\partial}{\partial y}\mathbf{e}_y$$

$$\mathbf{j} = -M(\phi)\nabla\mu \qquad \nabla\mu = \frac{\partial\mu}{\partial x}\mathbf{e}_x + \frac{\partial\mu}{\partial y}\mathbf{e}_y$$

$$\frac{\partial\phi}{\partial t} + \nabla\cdot\mathbf{j} = 0 \qquad \nabla\cdot\mathbf{j} = \left(\frac{\partial}{\partial x}\mathbf{e}_x + \frac{\partial}{\partial y}\mathbf{e}_y\right)\cdot(j_x\mathbf{e}_x + j_y\mathbf{e}_y) = \frac{\partial j_x}{\partial x} + \frac{\partial j_y}{\partial y}$$

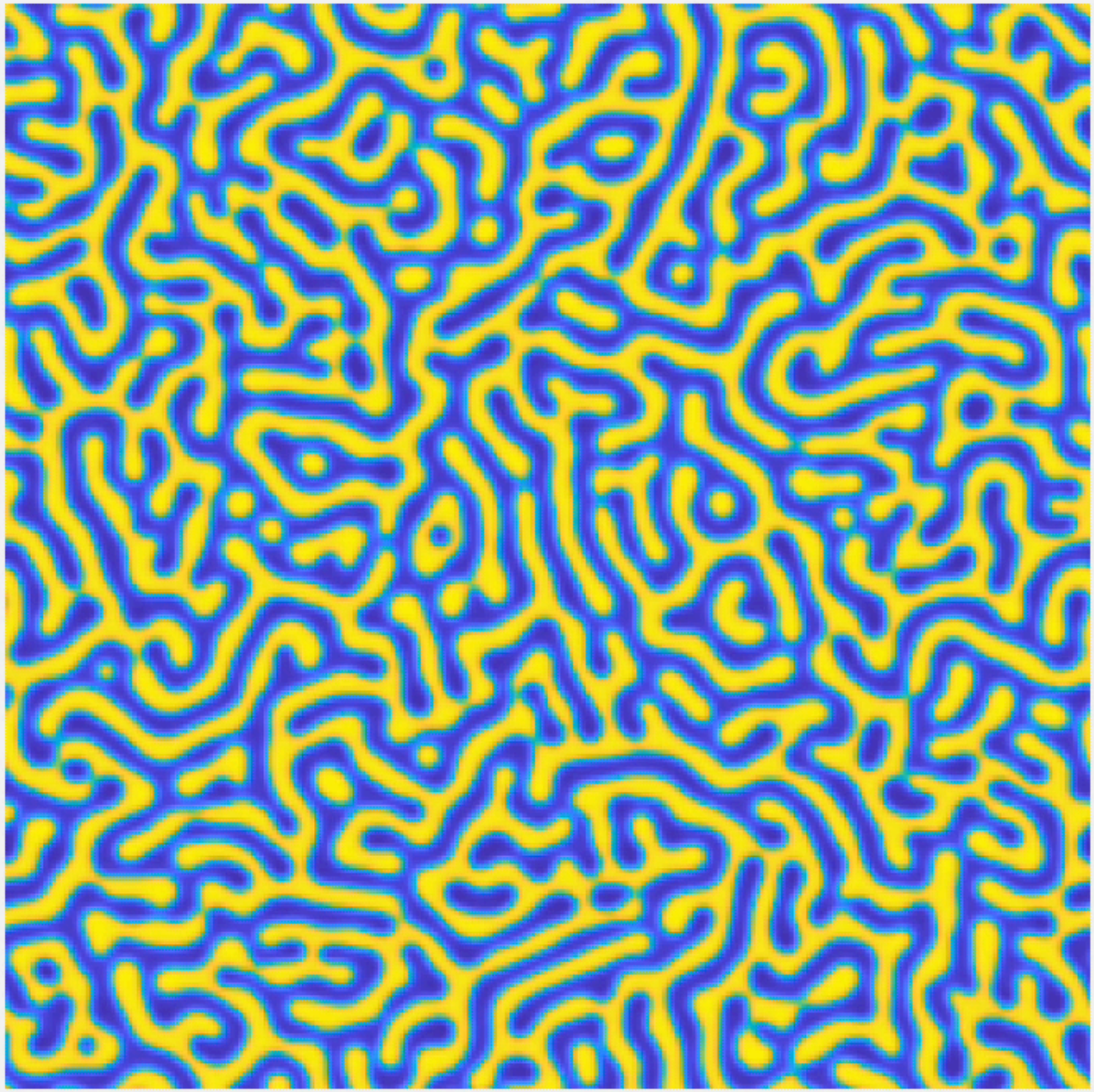
$$F[\phi] = \int\left(\frac{\sigma}{2}|\nabla\phi|^2 + \frac{1}{4}(\phi^2-1)^2\right)\,d\mathbf{x} \qquad |\nabla\phi|^2 = \left(\frac{\partial\phi_x}{\partial x}\right)^2 + \left(\frac{\partial\phi_y}{\partial y}\right)^2$$

$$\frac{\partial\phi}{\partial t} = M\nabla^2\left[\phi^3 - \phi - \sigma\nabla^2\phi\right]$$

$$\nabla^2\phi = \nabla\cdot\nabla\phi = \left(\frac{\partial}{\partial x}\mathbf{e}_x + \frac{\partial}{\partial y}\mathbf{e}_y\right)\cdot\left(\frac{\partial\phi}{\partial x}\mathbf{e}_x + \frac{\partial\phi}{\partial y}\mathbf{e}_y\right) = \frac{\partial^2\phi_x}{\partial x^2} + \frac{\partial^2\phi_y}{\partial y^2}$$

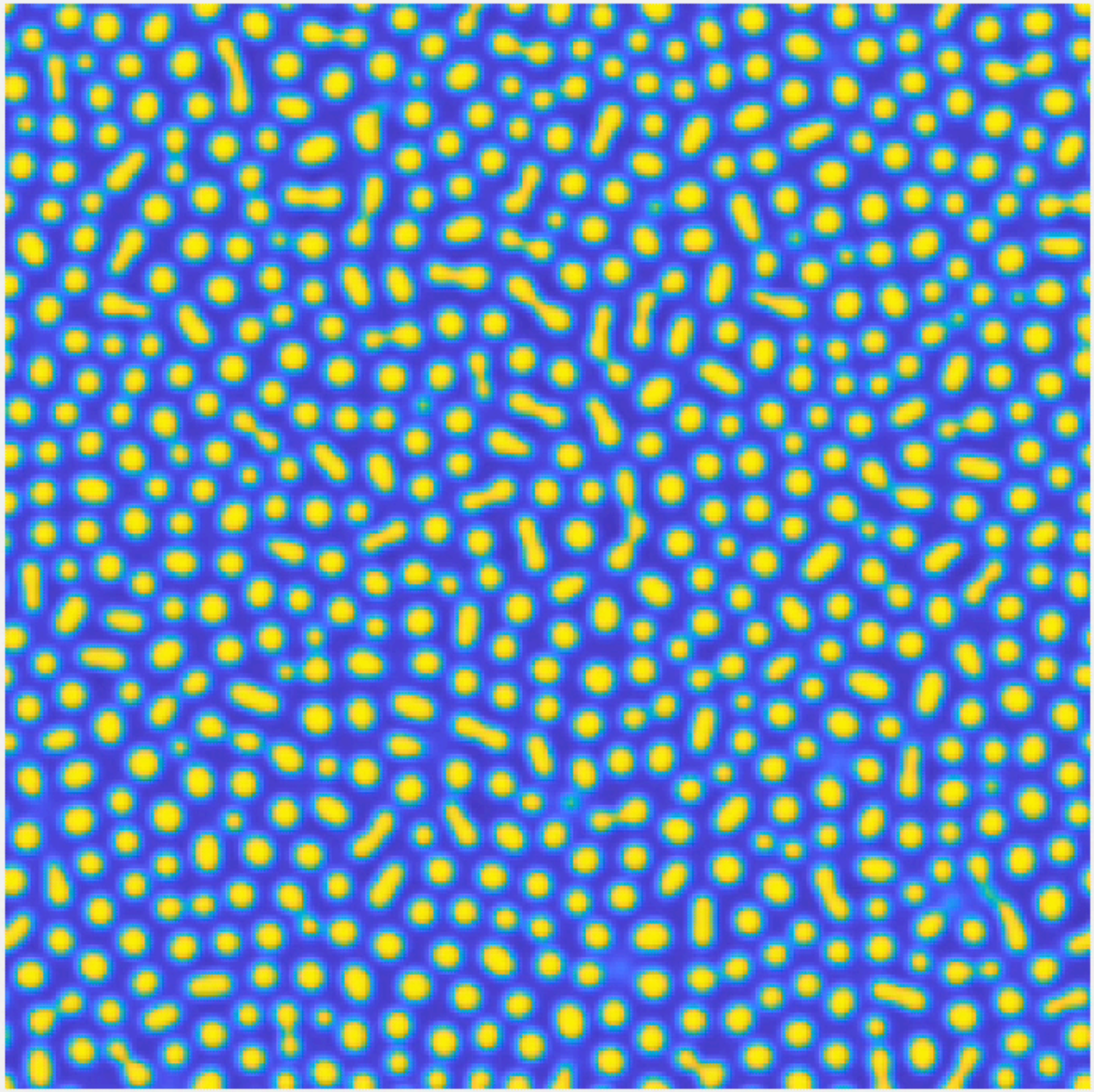
Cahn-Hilliard Phase-Field Simulation Time: 10.00

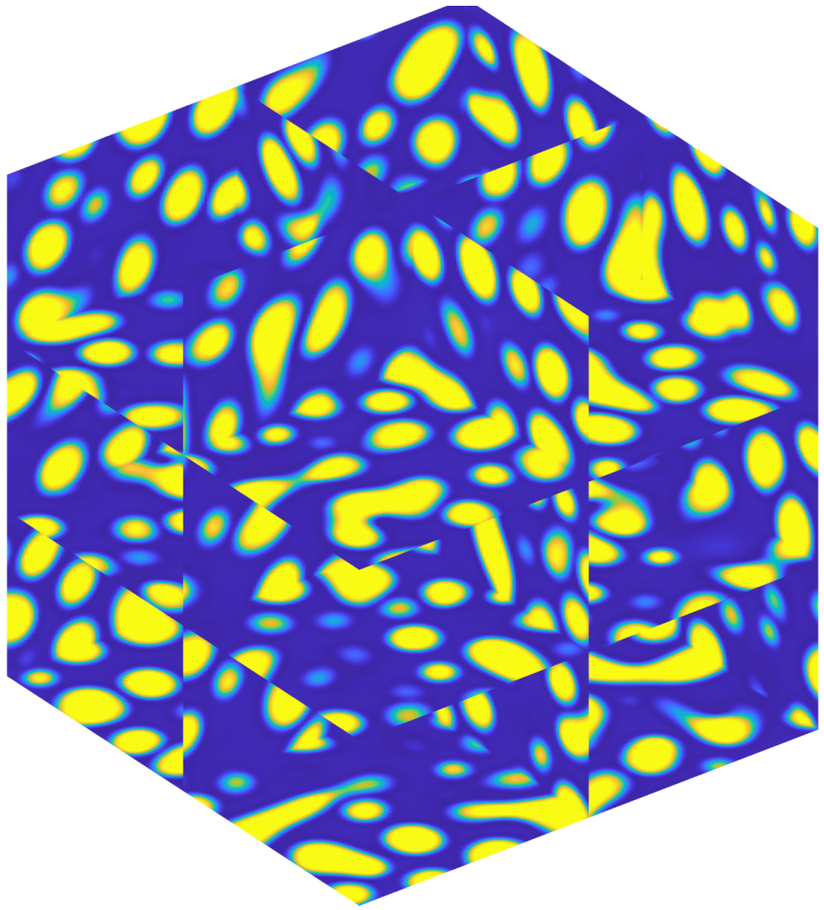
ic-random-bias0-kappa0-gamma0

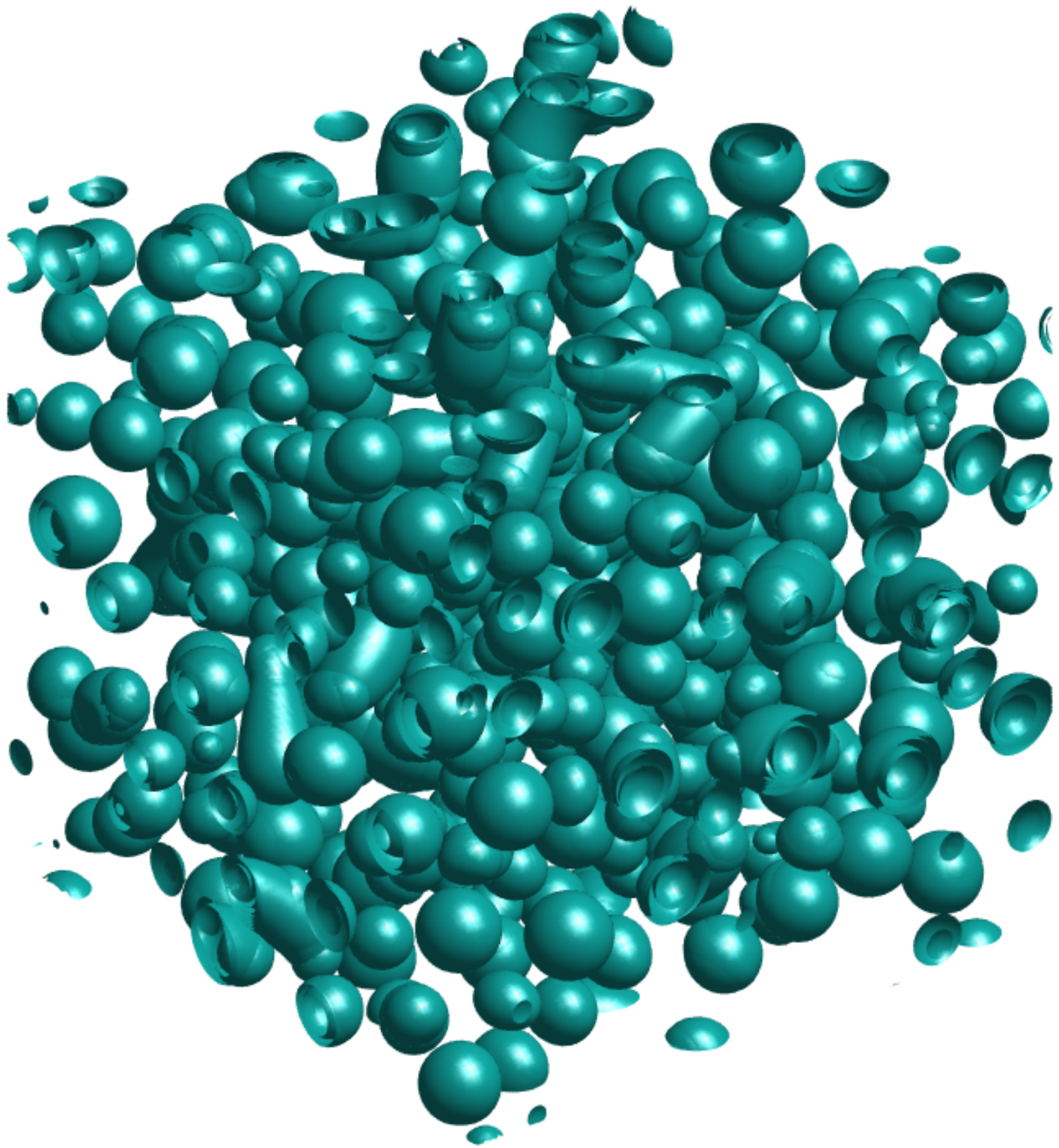


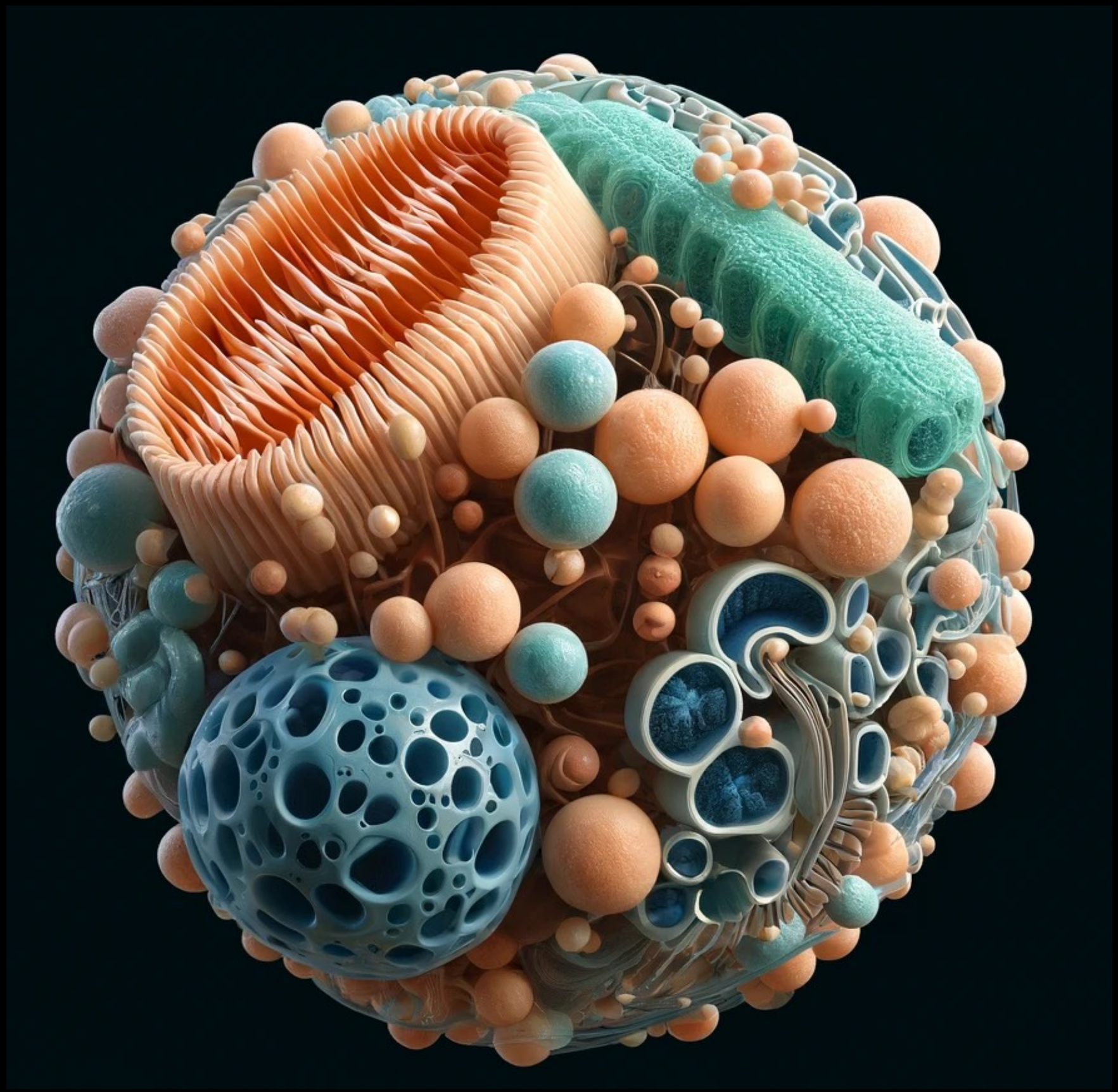
Cahn-Hilliard Phase-Field Simulation Time: 10.00

ic-random-bias-0.3-kappa0-gamma0









A **function** is a mathematical object that maps elements from one set (called the domain) to another set (called the codomain). For example, a function $f(x)$ takes a number x and returns another number $f(x)$.

A **functional**, on the other hand, is a specific type of mapping, but instead of taking a number as input, it takes an entire function as its input and returns a scalar (often a real number). Functionals are common in fields like calculus of variations and quantum mechanics.

Key differences:

- A **function** takes a variable as input and returns a value: $f(x)$.
- A **functional** takes a function as input and returns a scalar: $F[f]$.

For example, in physics, an energy functional might take a function that describes the configuration of a system and return the system's total energy, while a regular function could describe the potential energy at a specific point.

Suppose you have a functional $F[\phi]$, typically of the form:

$$F[\phi] = \int L(\phi, \nabla\phi, \ldots)\, d\mathbf{x}$$

where L is some function that depends on $\phi(x)$, its derivatives $\nabla\phi(x)$, and potentially higher-order derivatives.

The functional (variational) derivative $\delta F[\phi]/\delta\phi(x)$ is defined via

$$\int \frac{\delta F[\phi]}{\delta\phi(x)}\, \eta(x)\, d\mathbf{x} = \lim_{\epsilon \to 0} \frac{F[\phi + \epsilon\eta] - F[\phi]}{\epsilon}$$

```
I am working with the Cahn-Hilliard equations and have a free
energy functional given by F[\phi] = sigma/2 | \nabla \phi |^2 +
f(\phi).  Please use variational calculus to calculate the free
energy \mu = \delta F[\phi] / \delta \phi. In the derivation use
the limit of ( F[\phi + \eps \eta]-F[\phi] )/\eps as \eps goes
to zero. Be explicit about any integration by parts performed.
```

To compute the functional derivative $\delta F[\phi]/\delta\phi(x)$,

1. **Introduce a perturbation**: Let $\phi(x)$ be perturbed by a small amount $\epsilon\eta(x)$, where $\eta(x)$ is an arbitrary test function:

$$\phi(x) \to \phi(x) + \epsilon\eta(x)$$

   Substituting this into the functional, you get:

$$F[\phi + \epsilon\eta] = \int L(\phi + \epsilon\eta, \nabla(\phi + \epsilon\eta), \ldots)\, d\mathbf{x}$$

2. **Linearize the perturbation**: Expand $F[\phi + \epsilon\eta]$ in powers of ϵ up to first order:

$$F[\phi + \epsilon\eta] = F[\phi] + \epsilon \int \left(\frac{\partial L}{\partial \phi}\eta + \frac{\partial L}{\partial(\nabla\phi)} \cdot \nabla\eta + \cdots \right) d\mathbf{x} + O(\epsilon^2)$$

3. **Extract the first-order term**: The functional derivative $\delta F[\phi]/\delta\phi(x)$ is defined by the first-order term in ϵ. Therefore, collect terms that involve $\eta(x)$, and remove η to isolate the functional derivative.

Consider the Cahn-Hilliard free energy functional:

$$F[\phi] = \int \left(\frac{\sigma}{2} |\nabla \phi|^2 + f(\phi) \right) d\mathbf{x}$$

We want to compute the functional derivative $\delta F[\phi]/\delta \phi(x)$. Let $\phi(x)$ be perturbed as $\phi(x) \to \phi(x) + \epsilon \eta(x)$. Then, the perturbed functional is:

$$F[\phi + \epsilon\eta] = \int \left(\frac{\sigma}{2} |\nabla(\phi + \epsilon\eta)|^2 + f(\phi + \epsilon\eta) \right) d\mathbf{x}$$

Expand both terms in the integrand to first order in ϵ.

- **First term: Gradient term $\frac{\sigma}{2}|\nabla\phi|^2$**

  $$|\nabla(\phi + \epsilon\eta)|^2 = |\nabla\phi|^2 + 2\epsilon \nabla\phi \cdot \nabla\eta + O(\epsilon^2)$$

  The first-order term is $2\epsilon \nabla\phi \cdot \nabla\eta$.

- **Second term: Bulk energy term $f(\phi)$**

  $$f(\phi + \epsilon\eta) = f(\phi) + \epsilon \frac{\partial f}{\partial \phi} \eta + O(\epsilon^2)$$

  The first-order term is $\epsilon \frac{\partial f}{\partial \phi} \eta$.

Thus, the first-order variation of $F[\phi]$ is:

$$F[\phi + \epsilon\eta] - F[\phi] = \epsilon \int \left(\sigma \nabla\phi \cdot \nabla\eta + \frac{\partial f}{\partial \phi} \eta \right) d\mathbf{x}$$

To isolate $\eta(x)$, we remove the gradient on η by integrating by parts:

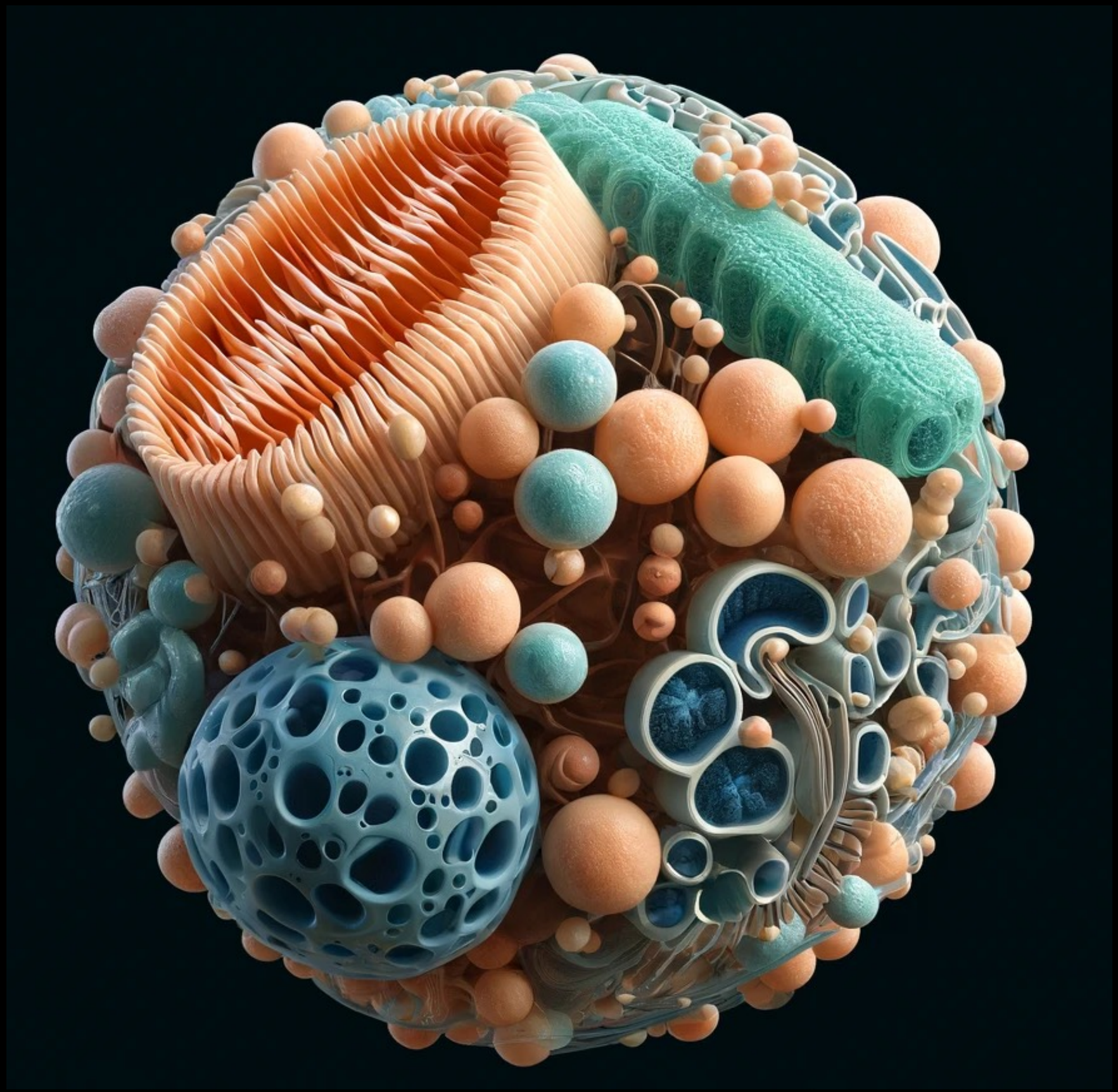
$$\int \sigma \nabla\phi \cdot \nabla\eta \, d\mathbf{x} = -\int \sigma \nabla^2 \phi \, \eta \, d\mathbf{x} \quad \text{(assuming boundary terms vanish)}$$

Now the first-order variation becomes:

$$F[\phi + \epsilon\eta] - F[\phi] = \epsilon \int \left(-\sigma \nabla^2 \phi + \frac{\partial f}{\partial \phi} \right) \eta \, d\mathbf{x}$$

Since $\eta(x)$ is arbitrary, we extract the functional derivative:

$$\frac{\delta F[\phi]}{\delta \phi(x)} = -\sigma \nabla^2 \phi + \frac{\partial f}{\partial \phi}$$



Numerical scheme – spectral method

After scaling time to eliminate the mobility M, the evolution equation for $\phi(\mathbf{x}, t)$ becomes

$$\frac{\partial \phi}{\partial t} = \nabla^2 \left[-\sigma \nabla^2 \phi + f(\phi) \right]$$

where $f(\phi) = (\phi^2 - 1)\phi$. The Fourier transform of this equation is **???**

Spatial Fourier transform

$$\hat{\phi}(\mathbf{k}, t) = \mathcal{F}(\phi(\mathbf{x}, t)) = \int_{\mathbb{R}^d} \phi(\mathbf{x}, t) e^{-i\mathbf{k}\cdot\mathbf{x}} \, d\mathbf{x}$$

Spatial inverse Fourier transform

$$\phi(\mathbf{x}, t) = \mathcal{F}^{-1}(\hat{\phi}(\mathbf{k}, t)) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \hat{\phi}(\mathbf{k}, t) e^{i\mathbf{k}\cdot\mathbf{x}} \, d\mathbf{k}$$

Fourier transform of the gradient

For a function $\phi(\mathbf{x})$ with its gradient $\nabla\phi(\mathbf{x})$, the Fourier transform of the gradient, $\mathcal{F}(\nabla\phi(\mathbf{x}))$, is given by:

$$\mathcal{F}(\nabla\phi(\mathbf{x})) = i\mathbf{k}\hat{\phi}(\mathbf{k})$$

where $\mathbf{k}$ is the wavevector in Fourier space, and $\hat{\phi}(\mathbf{k})$ is the Fourier transform of $\phi(\mathbf{x})$.

Fourier transform of the Laplacian

The Laplacian of a function $\phi(\mathbf{x})$ is given by:

$$\nabla^2\phi(\mathbf{x})$$

In Fourier space, the Laplacian becomes:

$$\mathcal{F}(\nabla^2\phi(\mathbf{x})) = -|\mathbf{k}|^2\hat{\phi}(\mathbf{k})$$

where $\mathbf{k}$ is the wavevector, and $\hat{\phi}(\mathbf{k})$ is the Fourier transform of $\phi(\mathbf{x})$.

Numerical scheme – spectral method $\quad \hat{\phi}(\mathbf{k}, t) = \mathcal{F}(\phi(\mathbf{x}, t))$

After scaling time to eliminate the mobility M, the evolution equation for $\phi(\mathbf{x}, t)$ becomes

$$\frac{\partial \phi}{\partial t} = \nabla^2 \left[-\sigma \nabla^2 \phi + f(\phi) \right]$$

where $f(\phi) = (\phi^2 - 1)\phi$. The Fourier transform of this equation is

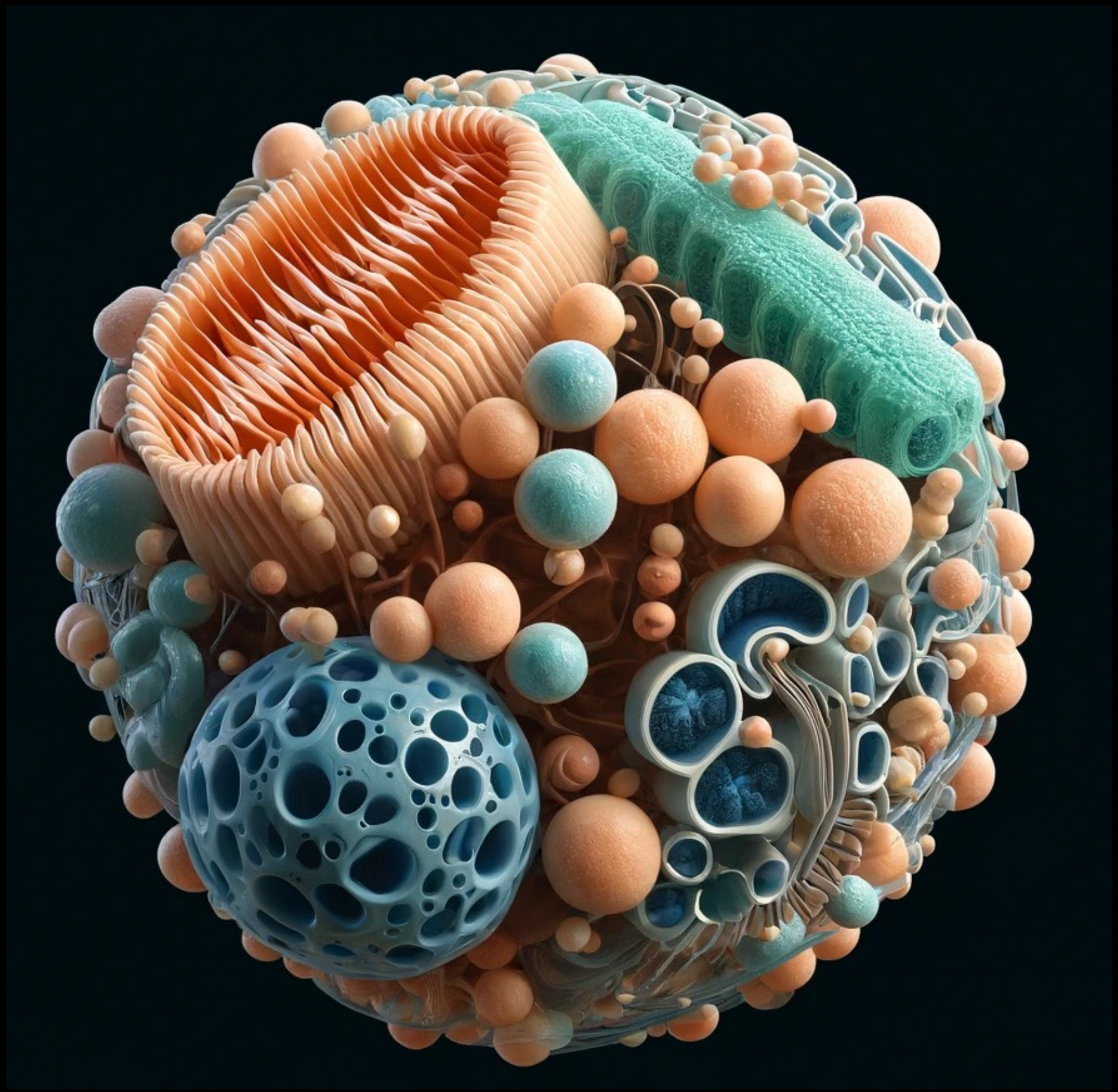
$$\frac{\partial \hat{\phi}}{\partial t} = -|\mathbf{k}|^2 \left[\sigma |\mathbf{k}|^2 \hat{\phi} + \hat{f} \right]$$

Derive semi-implicit finite difference scheme:

$$\hat{\phi}_{n+1} - \hat{\phi}_n = -\Delta t \left[\sigma |\mathbf{k}|^4 \hat{\phi}_{n+1} + |\mathbf{k}|^2 \hat{f}_n \right]$$

$$\hat{\phi}_{n+1} + \Delta t \sigma |\mathbf{k}|^4 \hat{\phi}_{n+1} = \hat{\phi}_n - \Delta t \, |\mathbf{k}|^2 \hat{f}_n$$

$$\hat{\phi}_{n+1} = \frac{\hat{\phi}_n - \Delta t \, |\mathbf{k}|^2 \hat{f}_n}{1 + \Delta t \, \sigma |\mathbf{k}|^4}$$



ChatGPT 4o

Are phase-field equations used to model membrane dynamics and/or membraneless organelles?

Yes, phase-field equations have been employed to model both membrane dynamics and membrane-less organelles (MLOs) via liquid-liquid phase separation (LLPS).

1. **Membrane Dynamics**: Phase-field models are particularly useful for simulating the dynamics of cellular membranes, as they can describe the evolution of complex shapes and topological changes, such as membrane fusion or fission, without needing explicit tracking of boundaries. The phase-field method represents the membrane as a diffuse interface between two regions (inside and outside the cell, for example), and its dynamics are governed by partial differential equations that include energy terms like surface tension and bending rigidity.
2. **Membrane-less Organelles (MLOs)**: Phase-field models are also applied to study the formation and dynamics of MLOs, which arise through LLPS. These models capture the spontaneous segregation of proteins and nucleic acids into dense and dilute phases, often representing the concentration of key components as a continuous field. The evolution of this field over time can be governed by Cahn-Hilliard-type equations, which describe phase separation driven by free energy minimization.

In both contexts, phase-field models allow the study of how complex structures and interfaces evolve dynamically, making them suitable for exploring processes like membrane deformations, fusion events, and the formation of organelles through phase separation.

Please explain how the Cahn-Hilliard equations are used to model LLPS

The Cahn-Hilliard equations are commonly used to model **liquid-liquid phase separation (LLPS)**, particularly in biological contexts like membrane-less organelles. These equations describe how two components (e.g., proteins and solvent, or two different proteins) spontaneously separate into distinct phases—dense and dilute—over time.

Key Components of the Cahn-Hilliard Model:

1. **Order Parameter (ϕ):**
    - The core idea is to define a scalar field, usually denoted as $\phi(\mathbf{r}, t)$, which represents the local concentration of one component (e.g., a protein) at position $\mathbf{r}$ and time t.
    - ϕ typically takes values near +1 in regions rich in the component of interest (dense phase) and values near -1 in regions depleted of it (dilute phase). In the interfacial region, ϕ varies smoothly between these extremes.
2. **Free Energy Functional:**
    - The behavior of the system is governed by a free energy functional $F[\phi]$, which drives the phase separation process. The free energy is typically written as:

$$F[\phi] = \int \left(\frac{1}{2} |\nabla \phi|^2 + f(\phi) \right) dV$$

    - The first term, $\frac{1}{2}|\nabla \phi|^2$, penalizes sharp gradients in ϕ, ensuring smooth interfaces between the phases.
    - The second term, $f(\phi)$, is a double-well potential (e.g., $f(\phi) = \frac{a}{4}(\phi^2 - 1)^2$) that favors segregation into the two stable phases (dense and dilute).

3. **Chemical Potential:**
    - The chemical potential μ, derived from the free energy, drives the dynamics. It is the functional derivative of the free energy with respect to ϕ:

$$\mu = \frac{\delta F[\phi]}{\delta \phi} = -\nabla^2 \phi + f'(\phi)$$

    - The chemical potential governs how the concentration ϕ changes in space and time.
4. **Cahn-Hilliard Equation:**
    - The time evolution of ϕ is governed by the Cahn-Hilliard equation:

$$\frac{\partial \phi}{\partial t} = \nabla \cdot (M(\phi) \nabla \mu)$$

    where $M(\phi)$ is the mobility, a parameter that describes how easily the components can diffuse.
    - This equation is a mass-conserving, diffusive process, where the concentration field ϕ evolves in such a way as to minimize the total free energy of the system.

Modeling LLPS with the Cahn-Hilliard Equation:

When used to model LLPS, the Cahn-Hilliard equation captures the spontaneous separation of a homogenous mixture into two distinct phases: a dense phase rich in certain biomolecules (e.g., proteins or RNA) and a dilute phase.

- **Nucleation:** Initially, small fluctuations in the concentration field ϕ occur due to thermal noise, leading to nucleation of dense regions.
- **Growth:** Once phase separation begins, domains of the dense phase grow over time at the expense of the dilute phase, eventually reaching a stable equilibrium.
- **Coarsening:** Larger dense domains grow as smaller ones shrink, driven by surface tension-like forces, in a process known as Ostwald ripening.

Applications in Membrane-less Organelles:

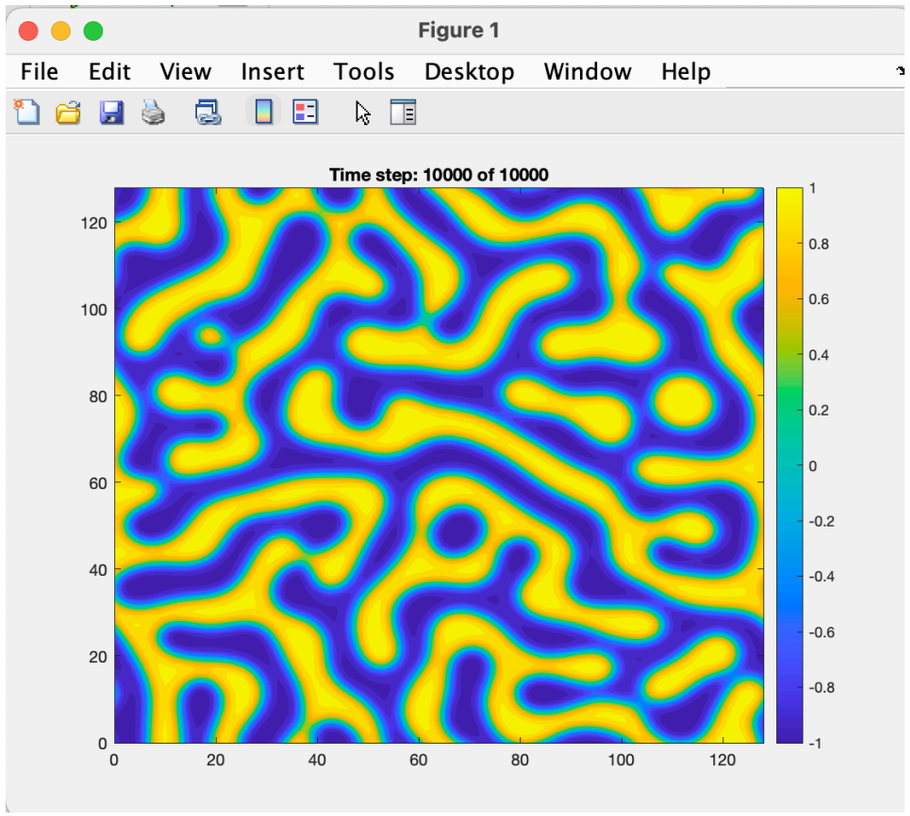
In biological systems, LLPS is central to the formation of membrane-less organelles like stress granules and the nucleolus. The Cahn-Hilliard equation can be adapted to study the kinetics of organelle formation, how environmental conditions (e.g., temperature, pH, or molecular crowding) affect LLPS, and how external forces like active processes (ATP-driven reactions) alter the equilibrium.

Key Features Captured:

- **Smooth Interfaces:** The diffuse nature of the phase-field representation captures smooth, continuous interfaces between the dense and dilute phases.
- **Dynamic Evolution:** The equation describes how concentrations evolve dynamically, including nucleation, growth, and coarsening of droplets.
- **Non-Equilibrium Phenomena:** The model can be extended to include non-equilibrium processes, like reactions or driven dynamics, to simulate biologically realistic conditions.

In summary, the Cahn-Hilliard equation is a versatile tool for modeling LLPS, capturing the essential physics of phase separation, from initial nucleation to the formation and stability of membrane-less organelles.

Please provide Matlab code that simulates the Cahn-Hilliard equation in two spatial dimensions.



MATLAB Code: Cahn-Hilliard Equation Using a Spectral Method

matlab Copy code

```
% Cahn-Hilliard Equation Simulation in 2D (Spectral Method)
% Parameters:
Nx = 128; % Number of grid points in the x direction
Ny = 128; % Number of grid points in the y direction
Lx = 128; % Length of the domain in x
Ly = 128; % Length of the domain in y
dx = Lx / Nx; % Grid spacing in x
dy = Ly / Ny; % Grid spacing in y
dt = 0.01; % Time step
M = 1.0; % Mobility
kappa = 1.0; % Interfacial energy coefficient
a = 1.0; % Coefficient in the free energy double-well potential

% Create the mesh grid
x = linspace(0, Lx, Nx);
y = linspace(0, Ly, Ny);
[X, Y] = meshgrid(x, y);

% Initial condition: random fluctuations around 0
phi = 0.5 * (rand(Nx, Ny) - 0.5);

% Define wave numbers for the Fourier transforms
kx = (2 * pi / Lx) * [0:(Nx/2-1), -Nx/2:-1]; % Wave numbers in x
ky = (2 * pi / Ly) * [0:(Ny/2-1), -Ny/2:-1]; % Wave numbers in y
[KX, KY] = meshgrid(kx, ky); % 2D wave number grid
k2 = KX.^2 + KY.^2; % Square of the wave number magnitude

% Time-stepping loop
tmax = 100; % Total simulation time
nsteps = round(tmax / dt); % Number of time steps
plot_interval = 100; % Plot every 100 time steps

% Fourier transform of the initial concentration field
phi_hat = fft2(phi);

% Simulation loop
for step = 1:nsteps
    % Transform phi back to real space to calculate the nonlinear term
    phi = real(ifft2(phi_hat));

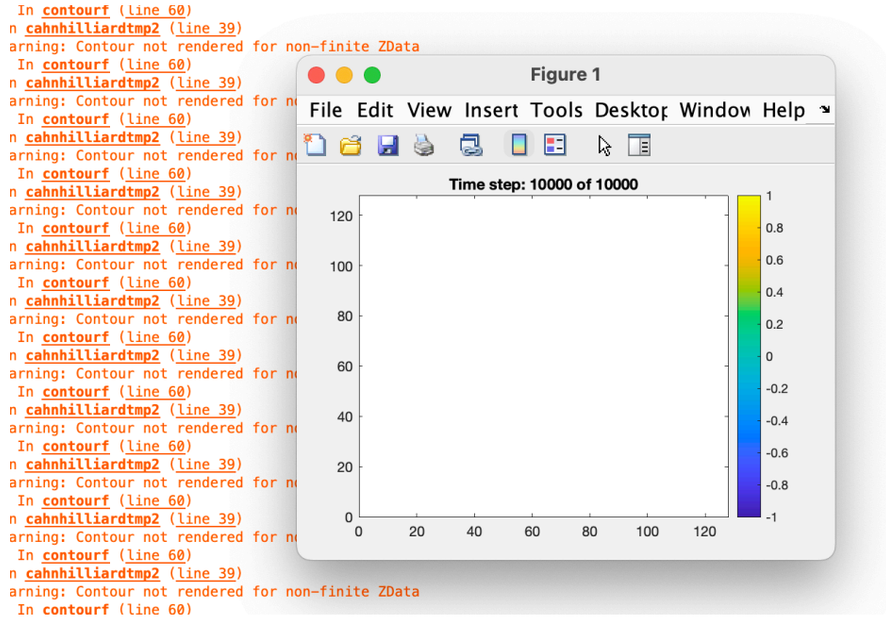
    % Calculate the chemical potential mu in real space
    mu = a * phi .* (phi.^2 - 1);

    % Transform mu to Fourier space
    mu_hat = fft2(mu);

    % Update phi_hat using the spectral form of the Cahn-Hilliard equation
    % dphi_hat/dt = - M * k^2 * (kappa * k^2 * phi_hat + mu_hat)
    phi_hat = (phi_hat - dt * M * k2 .* mu_hat) ./ (1 + dt * M * kappa * k2.^2);

    % Plot the concentration field every 'plot_interval' steps
    if mod(step, plot_interval) == 0
        phi = real(ifft2(phi_hat)); % Transform back to real space for plotting
        contourf(X, Y, phi, 20, 'LineColor', 'none');
        title(['Time step: ', num2str(step), ' of ', num2str(nsteps)]);
        colorbar;
        caxis([-1, 1]); % Limit color scale to match phase separation
        drawnow;
    end
end
```

I have heard that the Cahn-Hilliard equations can be extended so that their solution represents the dynamics of a biological membrane with spontaneous curvature. Please show how this would be done.



The Cahn-Hilliard equation can indeed be extended to model biological membranes, particularly when considering **spontaneous curvature**—an intrinsic property that drives membrane bending without the need for external forces. To incorporate spontaneous curvature into the Cahn-Hilliard framework, the free energy functional must be modified to account for membrane bending and curvature effects. Here's how this can be done:

1. Free Energy Functional with Curvature:

In the original Cahn-Hilliard model, the free energy functional includes a term to penalize sharp gradients (interface energy) and a double-well potential to describe phase separation. To model membrane dynamics with spontaneous curvature, we introduce a **Helfrich-like free energy** for membranes, which accounts for bending energy and spontaneous curvature. The modified free energy functional becomes:

$$F[\phi] = \int \left(\frac{\kappa}{2} (\nabla \phi)^2 + f(\phi) + \frac{\kappa_c}{2} (H - H_0)^2 \right) dV$$

```
% Chemical potential with curvature term
H = -k2 .* phi_hat; % Approximate curvature in Fourier space
mu = a * phi .* (phi.^2 - 1) + 0.5 * kappa_c * (H - H0).^2;
```

Where:

- κ is the interfacial energy coefficient (as in the classic Cahn-Hilliard model),
- $f(\phi)$ is the bulk energy potential (e.g., double-well potential),
- κ_c is the bending rigidity coefficient,
- H is the mean curvature of the membrane (which can be related to $\nabla \cdot \hat{n}$, where $\hat{n}$ is the unit normal to the membrane),
- H_0 is the spontaneous curvature.

That Matlab code does not work. More importantly, it seems to me that the quadratic spontaneous curvature term should be added to the free energy functional. The chemical potential would have a new term that was given by the functional derivative of this quadratic term.

You're absolutely right. The quadratic spontaneous curvature term should be added directly to the free energy functional, and its contribution to the chemical potential should be derived from the functional derivative of that term. Let's walk through how this should be done.

1. Free Energy Functional with Spontaneous Curvature:

To model spontaneous curvature, you extend the **Helfrich bending energy** to the phase-field framework. The free energy functional becomes:

$$F[\phi] = \int \left(\frac{\kappa}{2} (\nabla \phi)^2 + f(\phi) + \frac{\kappa_c}{2} (\nabla^2 \phi - H_0)^2 \right) dV$$

The **chemical potential** μ, which governs the evolution of ϕ, is given by the functional derivative of the free energy with respect to ϕ:

$$\mu = \frac{\delta F[\phi]}{\delta \phi}$$

The functional derivative yields:

$$\mu = -\kappa \nabla^2 \phi + f'(\phi) + \kappa_c \nabla^4 \phi - \kappa_c H_0 \nabla^2 \phi$$

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In cahnhilliardtmp3 (line 49)
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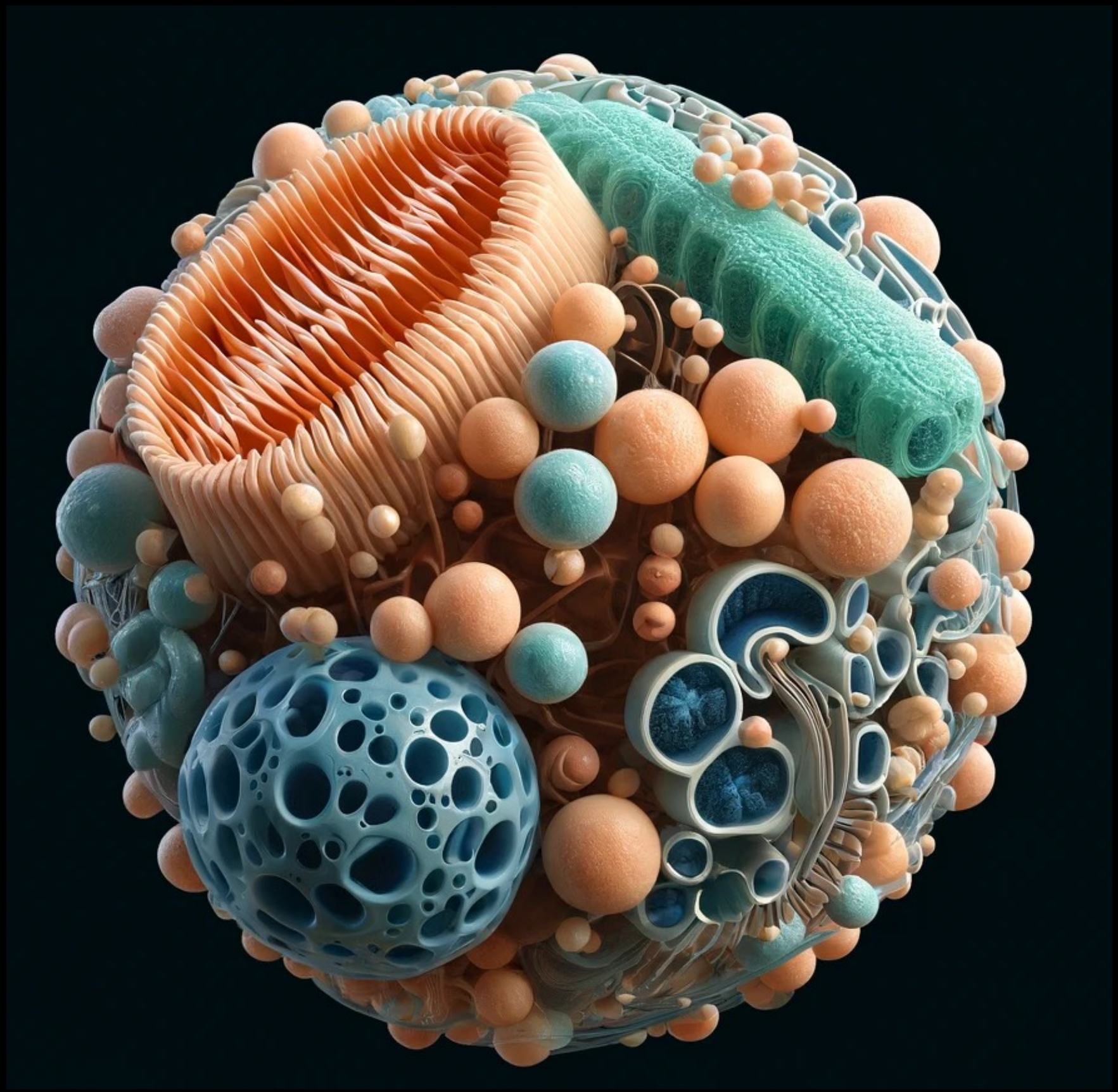
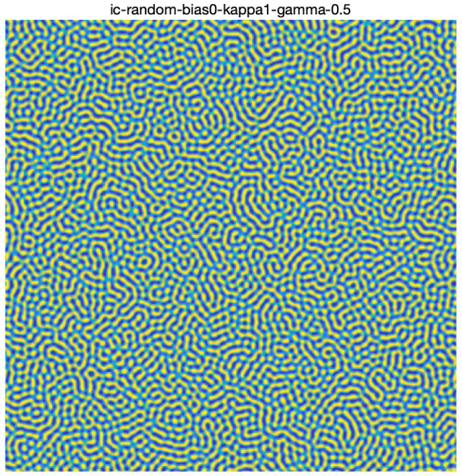
```

$$F_b[\phi] = \frac{\kappa}{2}(\nabla^2\phi - \gamma)^2$$

$$\mu_b = \frac{\delta F_b[\phi]}{\delta\phi} = \kappa\nabla^2(\nabla^2\phi - \gamma)d\mathbf{x}$$

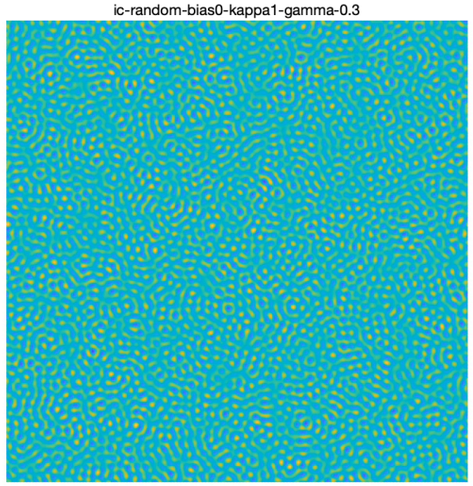
$$\mu_b = \frac{\delta F_b[\phi]}{\delta\phi} = \kappa\nabla^2(\nabla^2\phi - 2\gamma\phi)d\mathbf{x}$$





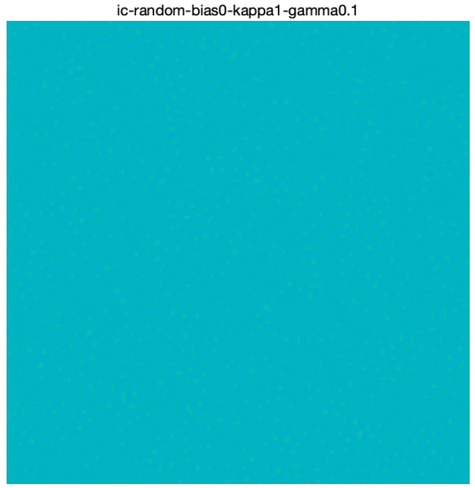
$$F_b[\phi] = \frac{\kappa}{2}(\nabla^2\phi - \gamma)^2$$

$$\kappa = 1$$

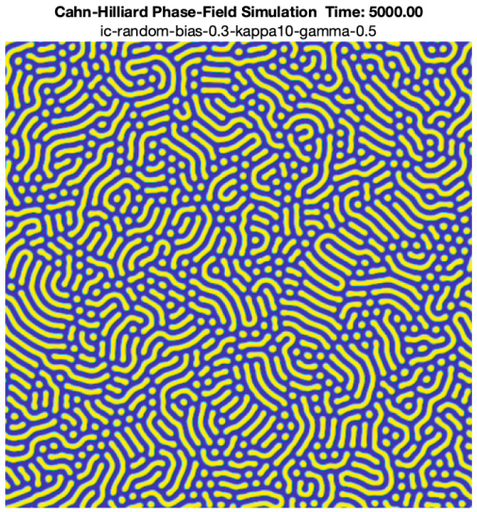


$\gamma < 0$

$\gamma = 0$

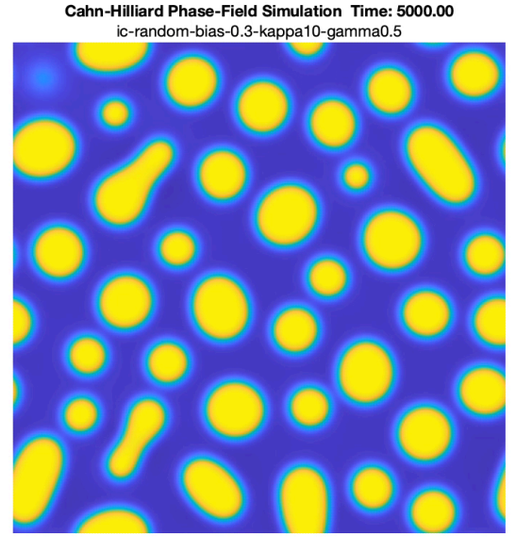


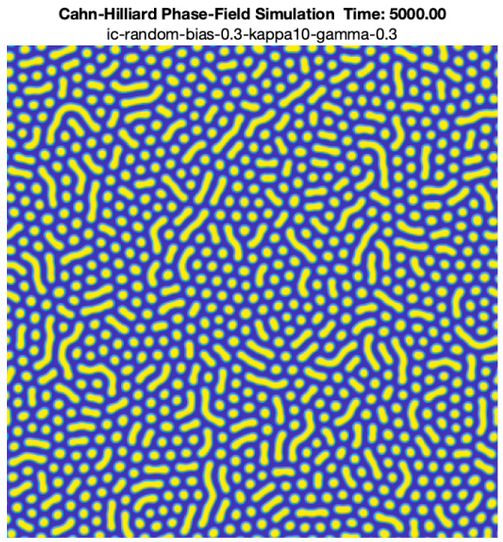
$\gamma > 0$

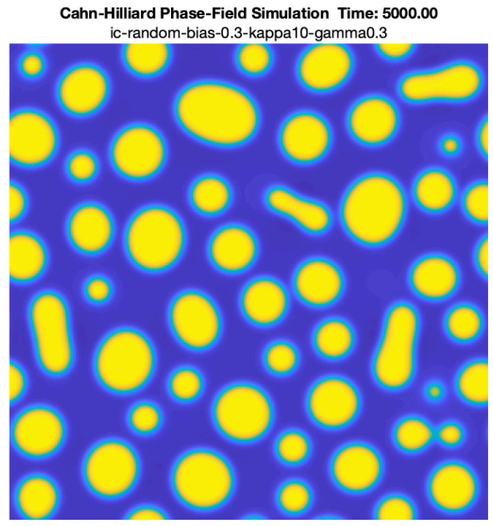


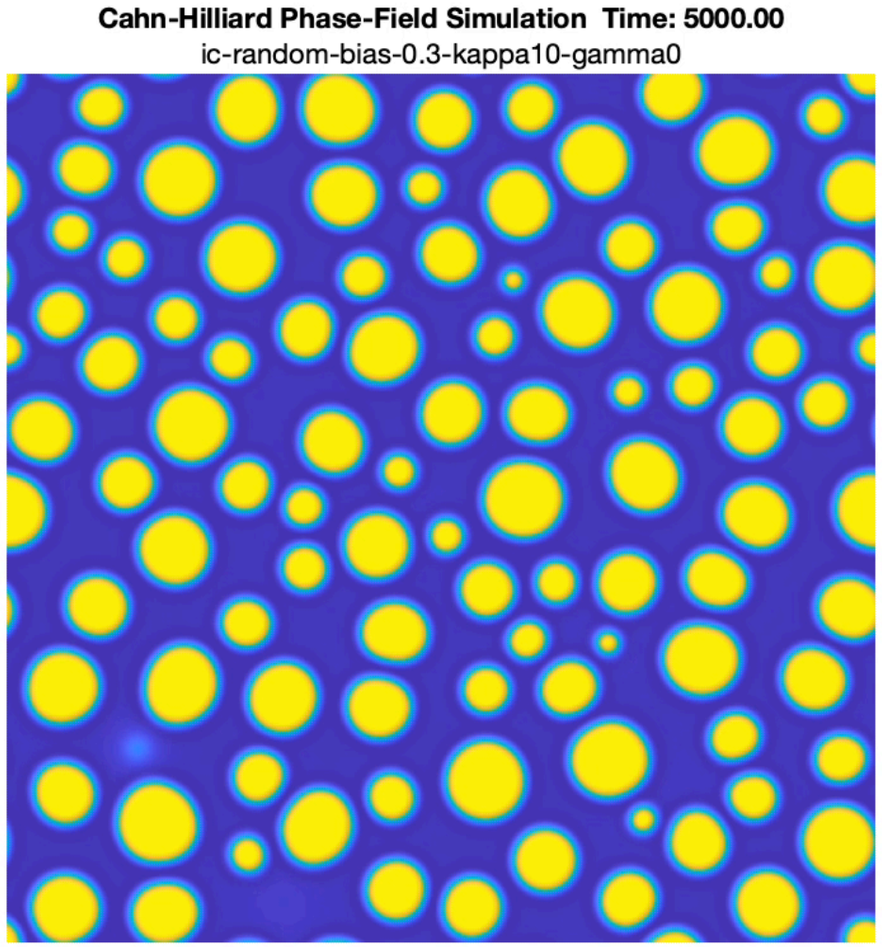
$$F_b[\phi] = \frac{\kappa}{2}(\nabla^2\phi - \gamma)^2$$

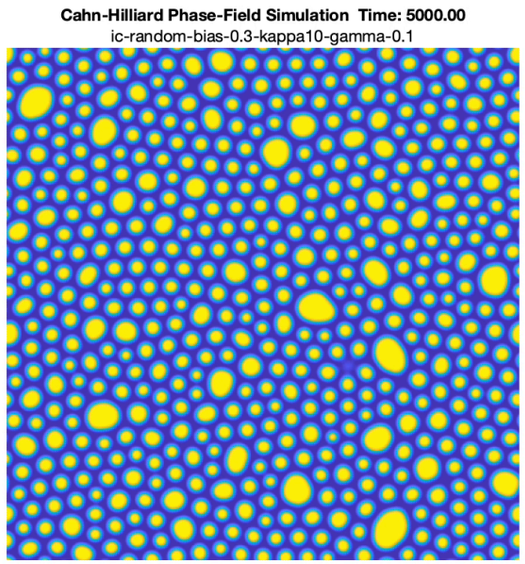
$$\kappa = 10$$

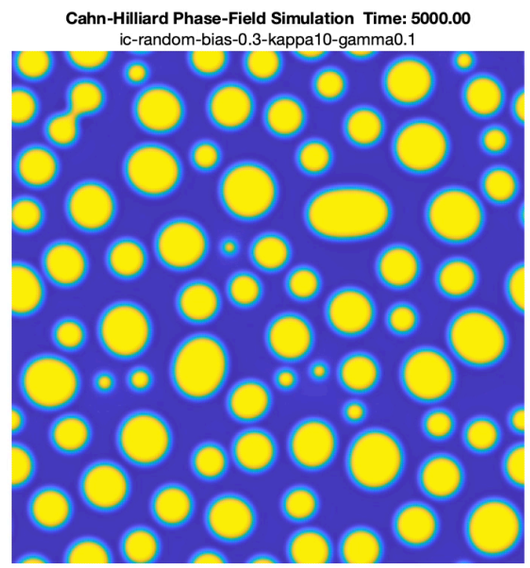








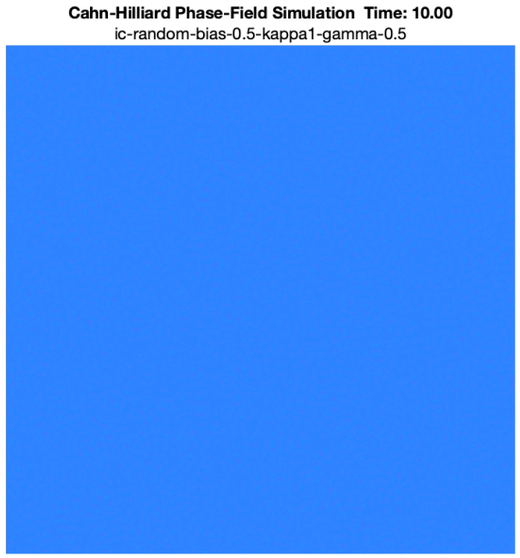




$\gamma < 0$

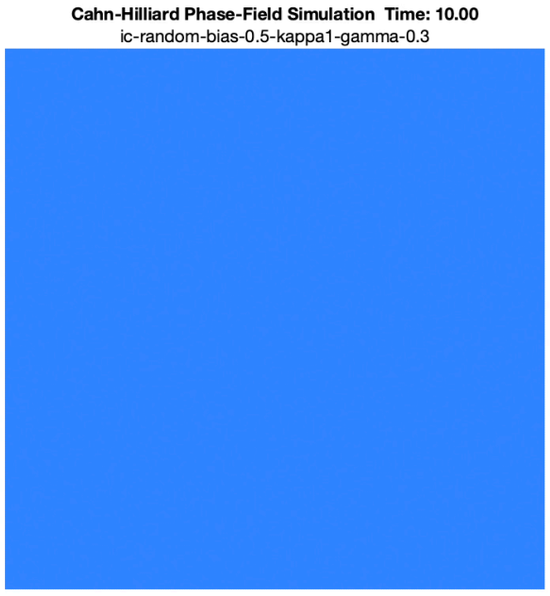
$\gamma = 0$

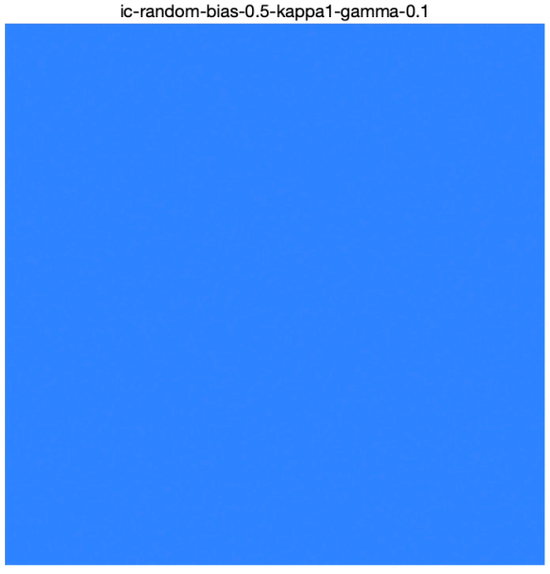
$\gamma > 0$



$$F_b[\phi] = \frac{\kappa}{2}(\nabla^2\phi - \gamma)^2$$

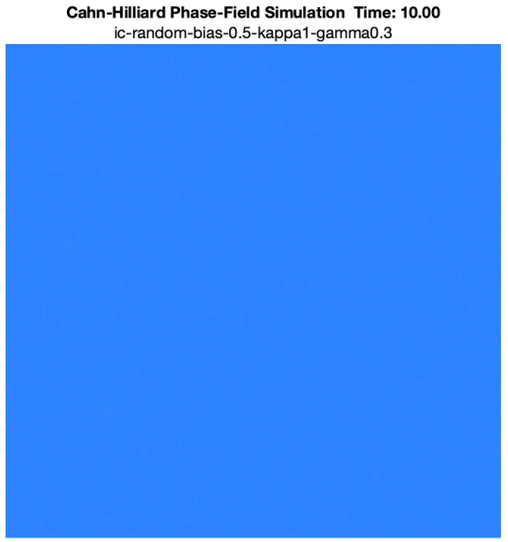
$$\kappa = 1$$

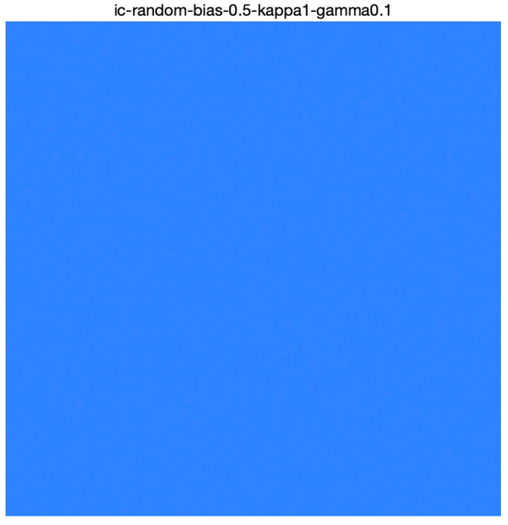




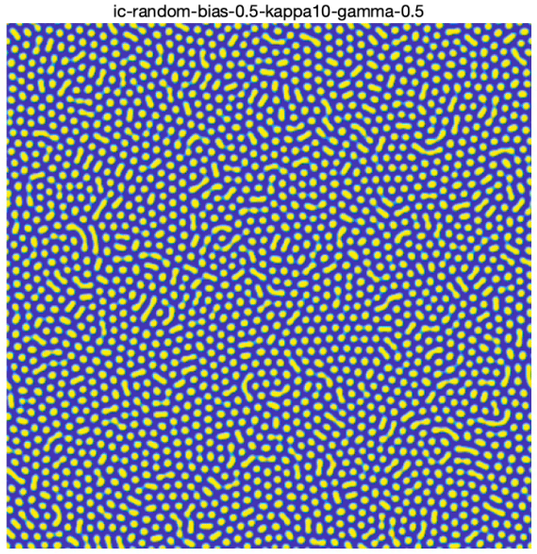
$\gamma < 0$

$\gamma = 0$



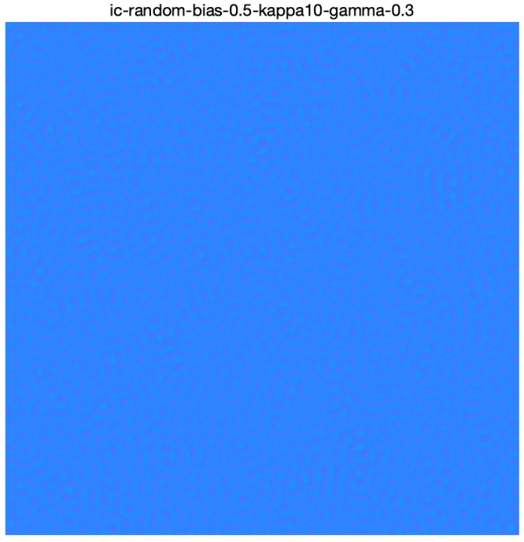


$\gamma > 0$



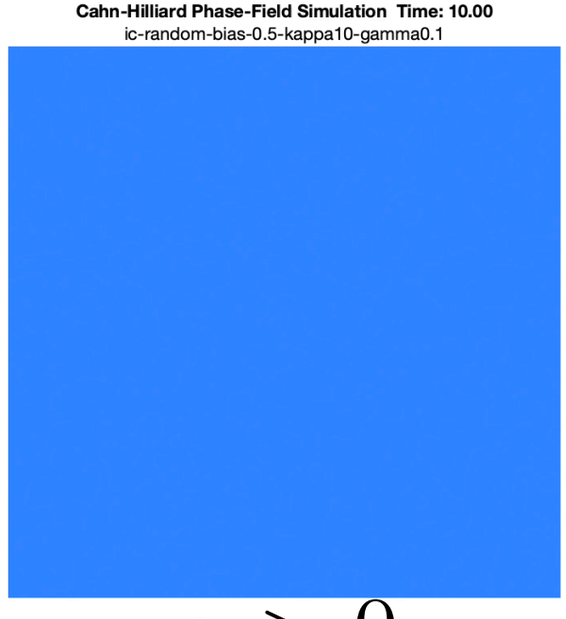
$$F_b[\phi] = \frac{\kappa}{2}(\nabla^2\phi - \gamma)^2$$

$$\kappa = 10$$

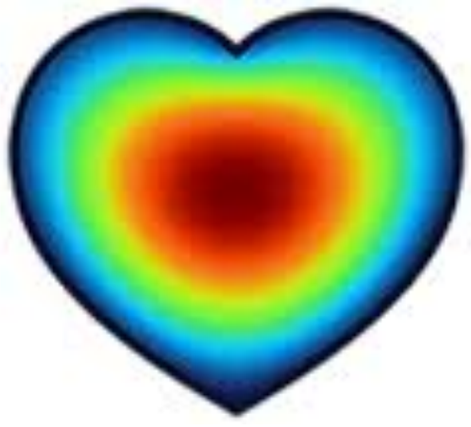


$\gamma < 0$

$\gamma = 0$



$\gamma > 0$



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